CHAPTER II.
METHODOLOGY

A. Project Initiation

a. Software Development Lifecycle

The development of TriOPS: Web-Based QR Production Process with Salary Management System and Workforce Analytics for Tagum Trio Lumber Corporation will follow the Modified Feature Driven Development (FDD) Software Development Life Cycle Model. FDD is a process for assisting teams in producing features incrementally that are useful for the end user. It is extremely iterative and collaborative in nature [5].

Table 1. Comparison of Software Development Models

Criteria	Spiral	Agile	FDD
Requirement Flexibility	Medium	High	High
User Involvement	Low	High	High
Development Approach	Risk-Driven Iterative	Incremental Sprints	Feature-by-Feature
Handling of Changes	Difficult	Flexible	Flexible
Risk Management	High	Medium	Medium
System Complexity Support	High	Medium	High
Continuous Feedback	Limited	Supported	Strongly Supported
Testing Approach	After Each Cycle	Every Sprint	Integrated Per Feature
Documentation Requirement	High	Minimal	Medium
Development Speed	Slower	Faster	Faster
Suitability for Modular System Design	Medium	High	High
Support for Parallel Feature Development	No	Partial	Yes

Based on Table , Feature Driven Development (FDD) was identified as the most suitable methodology for the development of the proposed TriOPS: Web-Based QR Production Process with Salary Management System and Workforce Analytics for Tagum Trio Lumber Corporation. Compared to the Spiral and Agile Models, FDD better supports feature-by-feature development, strong user involvement, and continuous feedback at every stage of the build process. These characteristics are essential in developing the interconnected modules of TriOPS, which include QR code-based production and attendance monitoring, salary component setup, basic pay and deduction computation, overtime and allowance calculations, final pay processing, workforce analytics, standard and custom report generation, and decision support dashboards.

However, since TriOPS also requires continuous validation of QR-scanned production data, real-time refinement of salary computation rules, and evolving workforce analytics aligned with lumber production operations, the proponents derived a Modified FDD Model by incorporating a continuous feedback loop and iterative improvement mechanism across all five phases of development Develop an Overall Model, Build a Feature List, Plan by Feature, Design by Feature, and Build by Feature. This modified approach allows the system to adapt to the dynamic operational requirements of Tagum Trio Lumber Corporation while ensuring that the workforce analytics components, such as production performance metrics, salary summaries, attendance trends remain accurate, reliable, and aligned with the day-to-day processes of the organization.

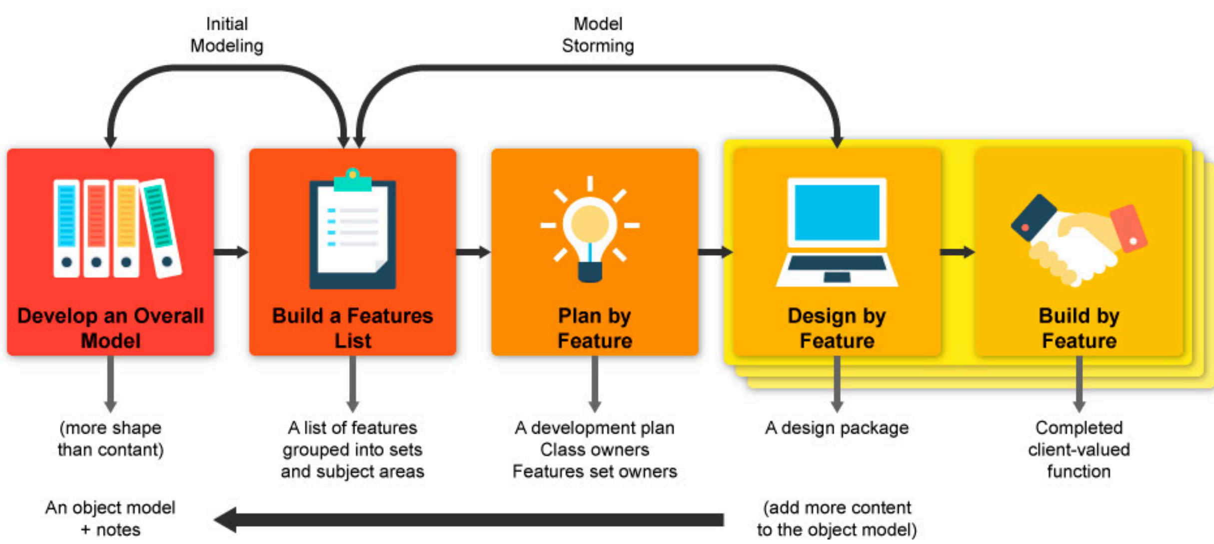


Figure 6. Feature Driven Development(FDD)

This study adopts the Feature-Driven Development (FDD) methodology in developing the *TriOPS: Web-Based QR Production Process with Salary Management System and Workforce Analytics*. FDD is an iterative and incremental software development approach that focuses on designing and building features that are valuable to the client. It is particularly suitable for this project because the system is composed of multiple modules such as QR-based production tracking, salary computation, employee management, and workforce analytics.

Feature-Driven Development (FDD) is better than other methods because it is simple, clear, and works fast. Unlike old ways that take too long or are hard to change, FDD builds small, useful parts one by one. Each part is finished in just 1 or 2 weeks, so you get working results quickly and can fix or change things easily. It uses words everyone understands, so business people and tech teams never get confused. It works well for both small and big teams, keeps work neat, and lets you know exactly when things will be done.

As shown in Figure 6, the Feature-Driven Development (FDD) model consists of five connected stages that guide the whole process step by step. The first stage is developing the overall model, where the team works with business owners to create a clear blueprint or structure of the entire system. This step maps out what the system needs to do, how different parts connect, and what data or rules are needed—such as how QR codes, employee records, salary rules, and reports will link together. It sets the full picture so everyone shares the same goal, prevents confusion, and gives a strong base for all the work that follows, without writing any code yet.

The second stage is building the feature list, where the big plan is broken down into small, useful pieces called features. Each feature is a single, clear function that brings real value, written in simple words everyone can understand like “generate one QR code,” “calculate basic pay”. This list acts as a complete checklist of everything that needs to be built, ensuring nothing important is missed, and helping both the business and technical teams agree on exactly what will be delivered.

Third comes planning by feature, which organizes when, how, and by whom each feature will be made. The team sorts features by priority, sets a clear timeline usually one to two weeks per feature and assigns tasks to people based on their skills. This step makes work predictable, so you know exactly when each part will be finished, how long it will take, and who is responsible. It avoids delays, stops work from overlapping, and keeps the whole project moving smoothly without chaos or confusion.

Next is designing by feature, where every single feature gets its own detailed plan before any work is done. For each item, the team draws how it will look, how it will work, what data it will use, and how it will connect to other parts of the system. For example, they plan exactly what information goes inside each QR code, how to create many at once, and how to avoid duplicates; or they map out the exact formulas and rules for tax and deductions based on local laws. This ensures every part is correct, safe, and fits perfectly, catches mistakes early, and keeps the whole system neat and easy to change later.

The final stage is building by feature, where every design is turned into real, working software. The team writes the code, tests each part carefully, and checks that it works exactly as planned and gives the right result such as creating correct and unique QR codes, calculating salaries accurately, or generating proper reports. Each feature is finished, tested, and ready to use one by one. This way, you get working results very fast, can fix small problems right away, and can easily add new functions or update rules whenever needed, without redoing the whole system from scratch.

The use of FDD ensures that the system is developed in a structured manner, where each feature is clearly defined, designed, and implemented in short development cycles. This allows continuous progress monitoring and ensures that each component of the system meets the required specifications.

b. Work Breakdown Structure

Project Milestone and Deliverables

According to Işık and Çifci (2023), milestones play a vital role in project timelines by designating important points of development and enabling monitoring of that progress. As shown in Figure 6, each stage of the FDD model marks a key project milestone and produces specific deliverables that move the work forward step by step, while still serving their core purpose, following clear implementation steps, and guaranteeing important results for the whole system.

The first stage is Develop Overall Model. Its milestone is Project Scope & System Blueprint Finalized, meaning the full structure and boundaries of the system are fully agreed and set. The deliverables here are system structure diagrams, process maps, data flow outlines, and a complete definition of all rules and connections—such as how employee data, QR code generation, salary computation, and workforce reports relate to each other. Its purpose is to establish a solid foundation and clear direction for everything that follows. Implementation involves joint meetings, mapping sessions, and discussions between the team and business owners to capture every requirement. This ensures that the whole system is built on a correct, shared understanding, that no part is missing or disconnected, and that all future work aligns perfectly with what the business truly needs.

The second stage is Build Feature List. Its milestone is Complete Feature Inventory Approved, confirming that every required function is listed and accepted by all sides. The deliverable is a detailed, prioritized feature list written in simple business terms including items like “Generate unique employee QR codes”, “Calculate monthly pay with deductions”, and “Produce workforce productivity reports”. Its purpose is to break the big system into small, clear, valuable pieces that everyone understands. Implementation happens by analyzing the blueprint, splitting big goals into single functions, and sorting them by importance. This ensures that every part built delivers real use, that nothing important is forgotten, and that there is no confusion about exactly what will be delivered in the end.

The third stage is Plan by Feature. Its milestone is Work Schedule & Resource Plan Ready, meaning the timeline, order, and responsible persons for all tasks are fixed. The deliverables are a development schedule, task assignment sheet, priority matrix, and estimated timelines for each feature (usually 1–2 weeks per item). Its purpose is to organize work so it flows smoothly, stays on time, and uses people and tools wisely. Implementation includes estimating effort, ranking features, resources are not wasted, deadlines are met, and the project progresses in a steady, controlled way.

The fourth stage is Design by Feature. Its milestone is Feature Specifications & Designs Approved, confirming that every function has a complete, correct plan before building starts. The deliverables are detailed design documents, data formats, logic flowcharts, formulas, and connection rules such as exact details embedded in QR codes, and how data moves between modules. Its purpose is to define exactly how each feature will work, look, and connect to others. Implementation involves designing each function separately, checking rules and compliance, and testing logic on paper or models. This ensures that every feature is accurate, follows laws and policies, has no errors, fits perfectly into the whole system, and is easy to change or update later.

The fifth stage is Build by Feature. Its milestone is Working Features Fully Tested & Released, marking that each function is finished, verified, and ready to use. The deliverables are working software modules, tested functions, user guides, and verified results like fully working QR code generator, salary processing tool, and workforce analytics dashboard. Its purpose is to turn all plans and designs into real, usable parts of the system. Implementation includes writing code, building the tool, running tests, checking outputs, and fixing issues immediately. This ensures that the final system works correctly, produces accurate and reliable results, is stable to use, and can be expanded or modified anytime without breaking existing functions.

Table 2. Project Milestone and Deliverables

Project Phases	Deliverables	Date
Develop an Overall Model	Business Goals and Expected Benefits Documentation	8-Feb-26
	System Purpose and Expected Benefits Documentation	13-Feb-26
	Stakeholder Interview Results	20-Feb-26
	Existing Processes and Documents Review	24-Feb-26
	User Needs and Functional Requirements Document	27-Feb-26
	Core Data Entities Defined (Employee, Attendance, Salary, Reports)	4-Mar-26
	Entity Relationships Established	10-Mar-26
	High-Level System Architecture Design	12-Mar-26
	Overall Project Model Reviewed and Approved	14-Mar-26
Build a Feature List	Attendance Management Features Identified	18-Mar-26
	Salary Calculation Features Identified	20-Mar-26
	Reporting and Analytics Features Identified	25-Mar-26
	Feature Purpose and Functionality Descriptions	26-Mar-26

	Feature Inputs, Outputs and Rules Specification	31-Mar-26
	Features Classified by Priority and Importance	3-Apr-26
	Validated Feature List Organized in Logical Order	8-Apr-26
	Scope and Boundaries Defined for Each Feature	10-Apr-26
	Effort and Time Estimates per Feature	13-Apr-26
	Required Resources and Skills Identified	17-Apr-26
	Feature Dependencies Documented	21-Apr-26
	Tasks Assigned to Team Members	7-May-26
	Project Schedule and Milestones Created	12-May-26
	Implementation Approach Outlined for Medium Priority Features	3-Jun-26
	Technical Requirements Determined	6-Jun-26
	Costs and Risks Estimated	12-Jun-26
	Delivery Timelines Set	17-Jun-26
	Feature Plans Aligned with Overall Project Plan	22-Jun-26
Plan by Feature	All Feature Plans Reviewed and Finalized	26-Jun-26
	User Interface Design for Attendance Recording	3-Jul-26
	Design Logic for Attendance Calculation	8-Jul-26
	Salary Structure and Rules Design	10-Jul-26
	Calculation Formulas and Conditions Design	14-Jul-26
	Basic Report Templates Design	16-Jul-26
	Data Storage and Database Structure Design	18-Jul-26
	Data Entry and Validation Rules Design	20-Jul-26
	Access Control and User Roles Design	22-Jul-26
	System Navigation and Flow Design	24-Jul-26
Design by Feature	Designs Checked Against Requirements and Standards	28-Jul-26
	QR Code Generation and Attendance Recording Feature	30-Jul-26
	Time-In and Time-Out Recording Logic	31-Jul-26
	Attendance Record Storage and Retrieval	3-Aug-26
	Attendance Summary View	5-Aug-26
	Salary Component Setup Interface	11-Aug-26
	Basic Pay and Deduction Calculations	13-Aug-26
	Overtime and Allowance Calculations	19-Aug-26
	Final Pay Computation Function	28-Aug-26
Build by Feature	Standard Report Generation	1-Sep-26

	Data Visualization and Charts	3-Sep-26
	Custom Report Creation and Export	4-Sep-26
	User Management and Access Control	4-Sep-26
	Data Backup and Maintenance Tools	9-Sep-26
	System Settings and Configuration	12-Sep-26
	Individual Feature Functionality Tested	19-Sep-26
	Features Validated Against Requirements	2-Oct-26
	Defects and Issues Documented	6-Oct-26
	System Integration Tested	8-Oct-26
	Data Flow and Consistency Verified	3-Nov-26
	System Compatibility and Performance Tested	6-Nov-26
	User Training Sessions Conducted	1-Feb-27
	User Manuals and Documentation Prepared	2-Mar-27
	Initial Technical Support Provided	5-Mar-27
	System Performance and Usage Monitored	9-Mar-27
	Regular Data Backups Performed	16-Mar-27
	Updates, Patches and Security Fixes Applied	18-Mar-27
	User Issues and Requests Addressed	25-Mar-27
	Maintenance Activities and Progress Documented	2-Apr-27

Integrated and Proposed New Processes

QR Code Recording Production Management

Figure 7 shows the proposed QR Code Recording Production Process. The process begins with the Leadman accessing the leadman page of the system, choosing the assigned production department, and accessing the list of every employee. Afterward, the Leadman requests the department details, then accepts and assigns each employee’s account to a specific production department. Once assigned, the system automatically stores the record of every employee in its database.

The Production Head monitors the list of employees per department, while the Leadman checks if the process is applicable for scanning. If applicable, the Leadman scans the finished product, and the system records this entry. If not applicable, the Leadman manually records the quantity or thickness of the product. The system then calculates the price of every finished product based on its fixed price. Meanwhile, employees can view the current produced products and total earnings, and the

Leadman can also view all previous recorded reports.At the end of the day, the system generates a daily report, and the Production Head keeps and saves this report for reference. This proposed process improves efficiency by automating recording and calculation, centralizing all production data, and providing clear reports, thereby reducing manual work and making production monitoring easier and more accurate.

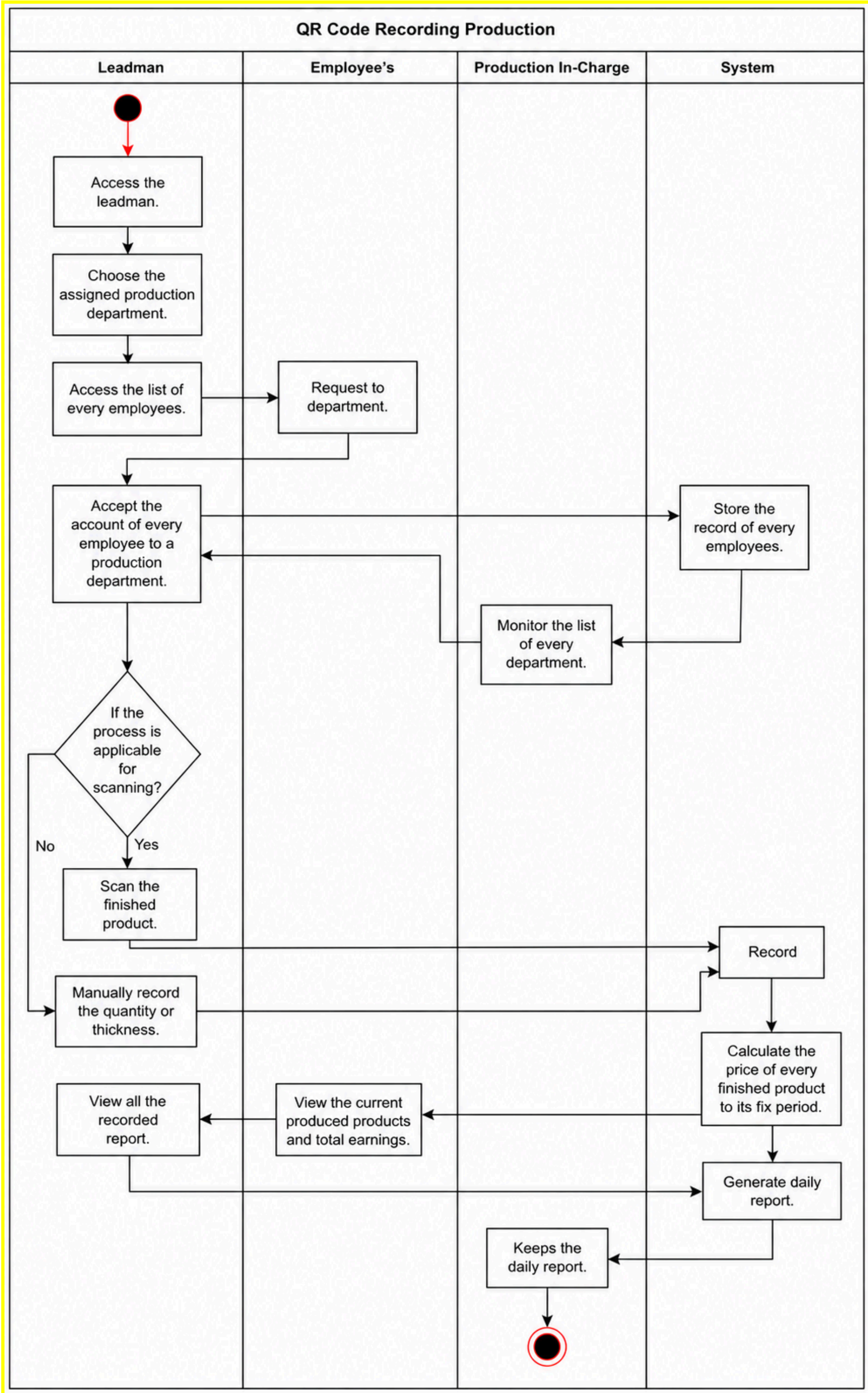

QR Code Recording Production
Leadman
Employee's
Production In-Charge
System
Access the leadman.
Choose the assigned production department.
Access the list of every employees.
Request to department.
Accept the account of every employee to a production department.
Store the record of every employees.
Monitor the list of every department.
If the process is applicable for scanning?
No
Yes
Scan the finished product.
Record
Manually record the quantity or thickness.
Calculate the price of every finished product to its fix period.
View all the recorded report.
View the current produced products and total earnings.
Generate daily report.
Keeps the daily report.

Figure 7. QR Code Recording Production Management (dili production head it should be Production In-Charge)

Leave Management Process

Figure 8 illustrates the proposed Leave Management Process. This workflow starts when the employee logs into the system and fills out the leave application form. Right after submission, the system automatically records all request details and forwards the application directly to the HR department for initial checking and review.

HR first reads and reviews the request letter, then checks and identifies how often the employee has filed for leave. If the application exceeds the maximum allowed leave limit, the process ends here, and the employee gets a notification about the result. If it is within the allowed limit, the request moves forward. The accounting staff translates the letter into the language preferred by the General Manager, then the system validates the document before sending it for final approval.

The General Manager then decides whether to approve or reject the request. If disapproved, the system sends a notification to the employee immediately. If approved, the system generates an official confirmation and updates the request status. Finally, the employee receives the notification and can view the latest status of their leave application within the system.This designed process makes leave requests simpler and faster. It organizes all steps clearly, reduces manual paperwork, ensures proper review and approval, and keeps the employee updated at every stage — making the whole leave management system more organized, transparent, and efficient.

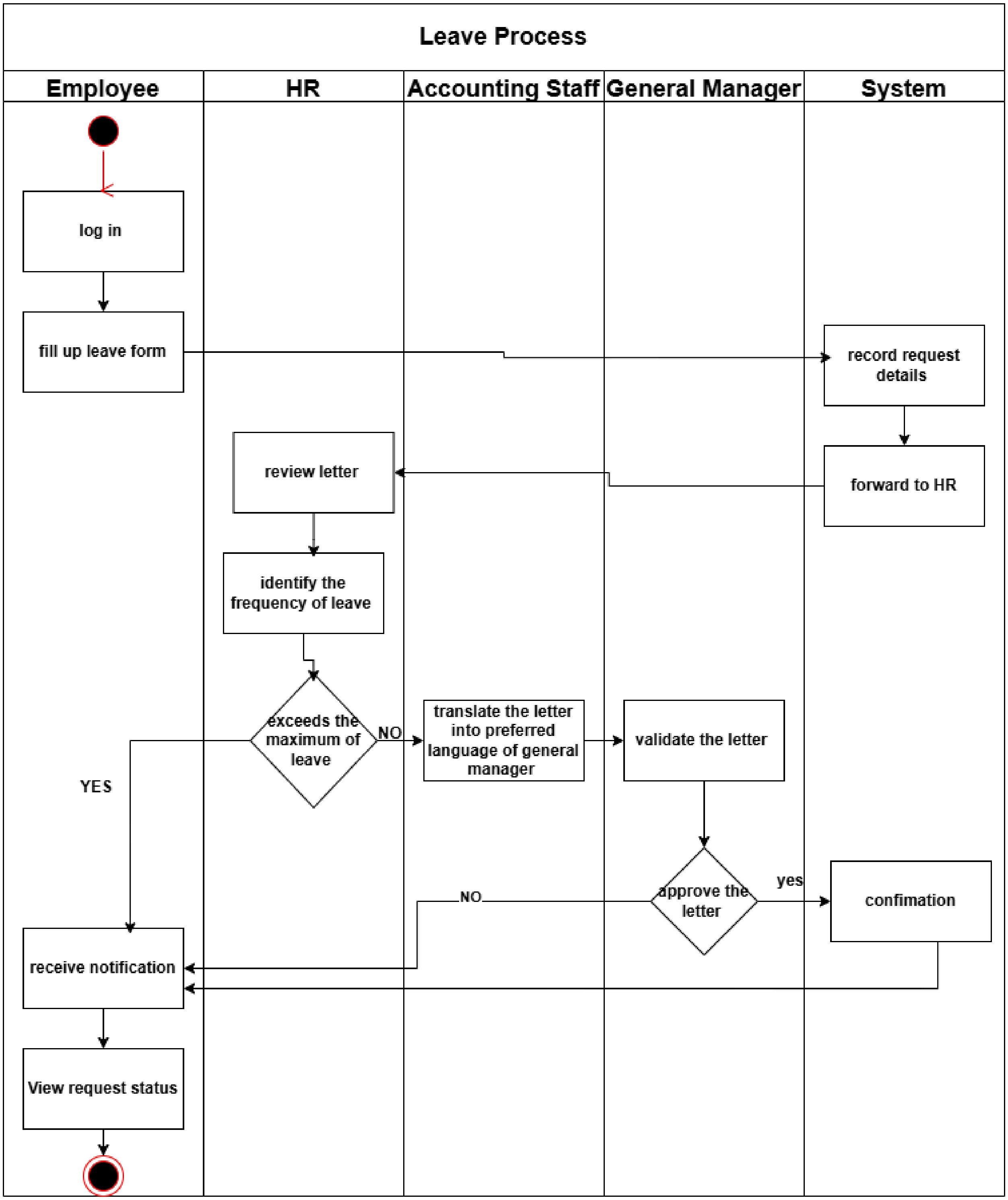


Figure 8. Leave Management Process

Salary Management Process

Figure 8 presents the proposed Salary Management Process, outlining the clear flow from data gathering to final salary release. The process begins when the Production Head logs into the system and opens the employee dashboard. Based on the data available, the system automatically computes the salary per department and generates the total salary amount for each employee.

Once computed, the HR or Accounting Head reviews the salary details and the generated report. After checking, they validate the report. If there are errors or corrections needed, the report is sent back for revision. If correct, it is forwarded to the General Manager for final assessment and approval. The General Manager reviews the full report and decides whether to approve or reject it. When approved, the team prepares the salary breakdown, and employees receive an official notification regarding their salary details. Employees may raise concerns if there are any issues; if there are none, they confirm and accept the amount. After confirmation, the system records whether the payment is made via cash or credit, then sends the record to HR or Accounting. Finally, the department releases the salary, marking the end of the process.

This structured workflow ensures accuracy, transparency, and proper monitoring at every stage. It reduces manual calculation errors, keeps all records organized, and makes salary computation and distribution systematic, reliable, and easy to track for all involved parties.

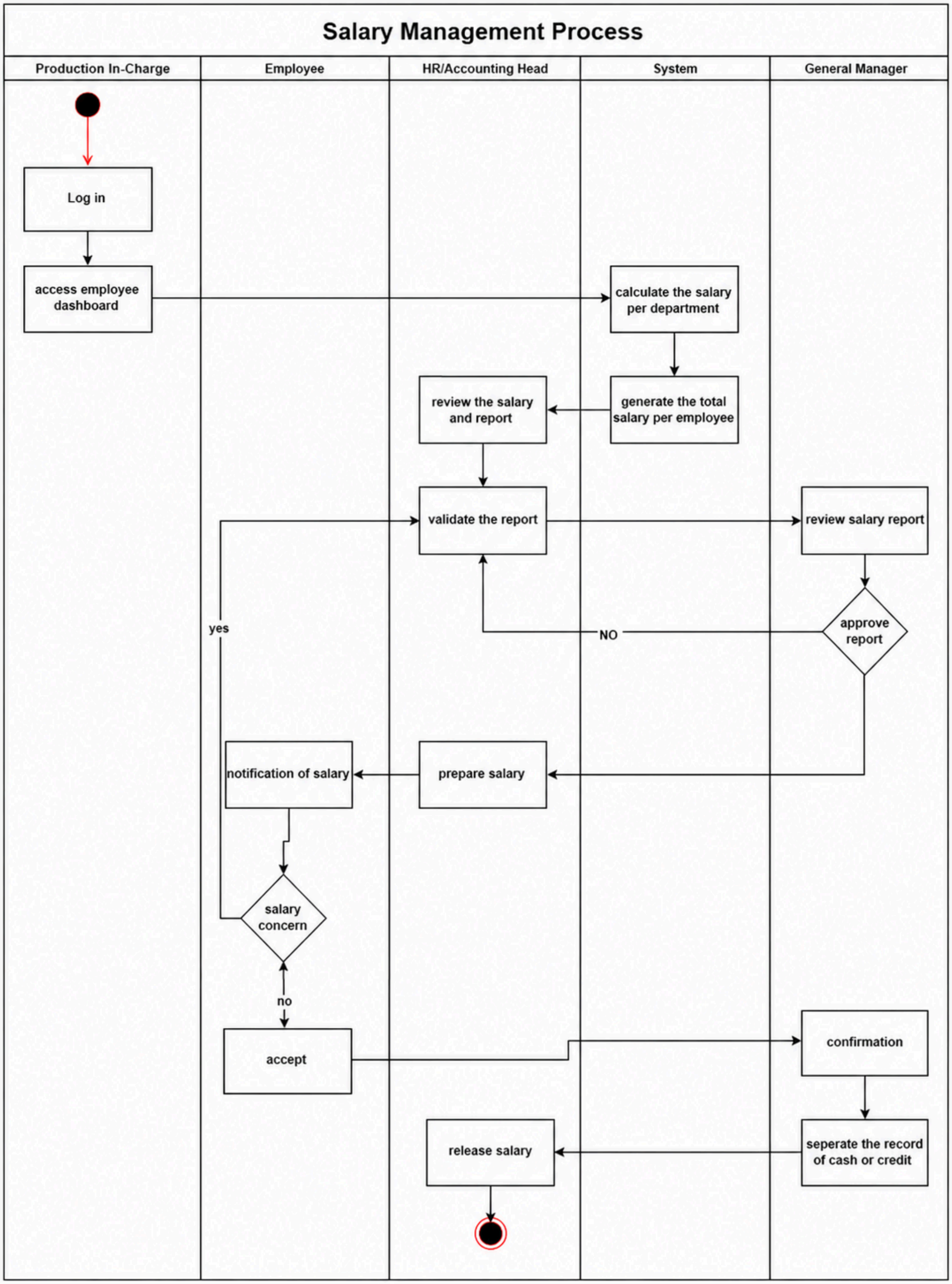

Salary Management Process
Production In-Charge
Employee
HR/Accounting Head
System
General Manager
Log in
access employee dashboard
calculate the salary per department
generate the total salary per employee
review the salary and report
validate the report
review salary report
approve report
NO
yes
notification of salary
prepare salary
salary concern
no
accept
confirmation
seperate the record of cash or credit
release salary

Figure 9. Salary Management Process(dili production head it should be Production In-Charge)

Reassign to the Production Department

Figure 10 illustrates the proposed process flow for Reassigning an Employee to a Production Department in TriOPS, showing the roles and responsibilities of each participant involved — the CEO/General Manager, the Employee, the Leadman, the System, and the Production In-Charge.

The process begins when the CEO/General Manager opens the Production Page and scouts the available production departments to assess workforce distribution. Upon identifying the department with the least number of employees, the CEO or manager decides to transfer an employee to balance the workload across production areas.

Once the transfer decision is made, the employee accesses their account in the system and submits a formal request for approval to transfer to the identified production department. At the same time, the Leadman of the receiving department receives a notification regarding the incoming employee and proceeds to accept the reassigned employee into their team.

Upon acceptance, the System takes over by recording the employee's daily production output and automatically calculating the employee's salary based on the number of finished products recorded — consistent with the piece rate computation method used by Tagum Trio Lumber Corporation.

On the side of the Production In-Charge, the reassigned employee is monitored throughout their stay in the new department. Once the monitoring period is completed, a decision is made on whether the employee should be returned to their original production department. If the answer is Yes, the employee is returned to their original placement and the system updates the records accordingly, completing the reassignment process.

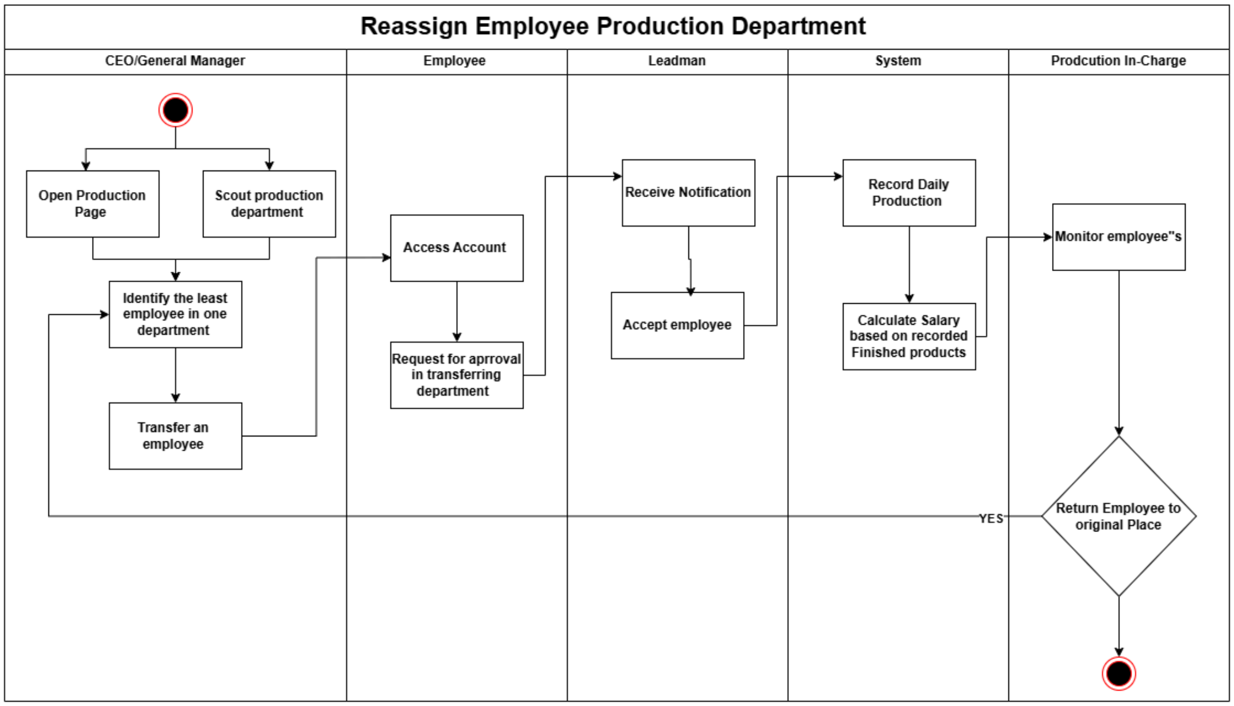


Figure 10. Reassign to the Production Department

c. Development Team

Tagum Trio Lumber Corporation has agreed to help the proponents in doing the research study. The company allows the proponents to do environmental scanning in its operations to collect the needed information for the system.The success of this project is further bolstered by the involvement of key individuals. Ms. Rhejane Austral, as the Production Head of the company, will serve as the Subject Matter Expert. Her deep operational knowledge will be invaluable in overseeing crucial aspects of quality and domain validity. She is expected to provide essential requirements based on real-world production processes and rigorously validate system functionalities, ensuring the proposed solution meets the company's stringent standards, enhances operational efficiency, and is fully embraced by end-users. Guiding the entire initiative as the Project Adviser is Ms. Monta Stacey Nicole Marie, whose academic and technical oversight will be pivotal in steering the development team through complex challenges, ensuring methodological soundness, and providing expert mentorship.

Under the leadership of Lance Benjamin as Project Manager, who will meticulously plan, execute, and oversee all project phases to ensure timely delivery within scope and budget. Ralph Zyron, serving as the System Analyst, will be responsible for translating the company's operational needs and Ms. Austral's insights into detailed system specifications and functional designs. John Kier Turco, as the Project Developer, will then take these designs to build, code, test, and implement the robust software solution. Along with other dedicated team members, the project is poised for smooth and efficient progress. The proposed system is specifically designed to significantly enhance current processes, accelerate operational workflows, and empower Tagum Trio Lumber Corporation with superior tools for comprehensive business management.

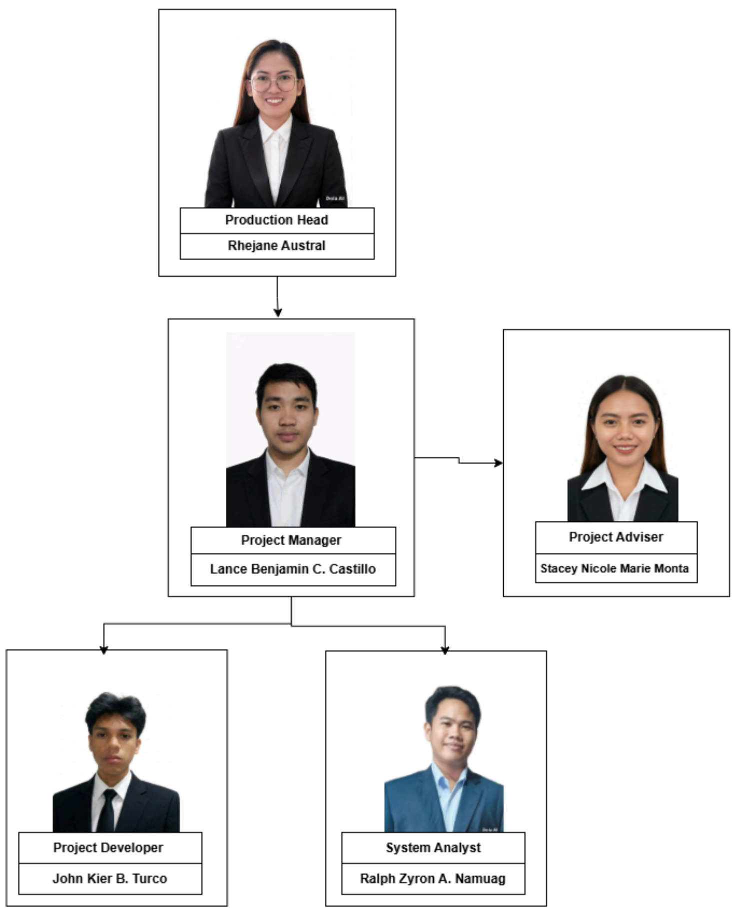


Figure 11. Development Team

d. Feasibility

Operational Feasibility

The organization is currently experiencing challenges related to manual processes in the recording, monitoring, and submission of production outputs, which result in delays, inconsistencies, and operational inefficiencies. The proposed system is operationally feasible as it directly addresses these concerns by digitizing and streamlining production-related processes. Through automation and centralized data management, the system is expected to improve accuracy, reduce paperwork, and enhance the efficiency of workflow within the production area.

The Production Personnel, as the primary users of the system, are anticipated to benefit from faster processing and more organized documentation of accomplished tasks. Since the system is developed based on the actual workflow of the organization, it aligns effectively with existing operational practices, thereby facilitating ease of adoption. Furthermore, management support reinforces the implementation of the system, as it aims to improve productivity and operational efficiency. The system is also designed with a user-friendly interface, ensuring that users can operate it effectively with minimal training or orientation. Overall, the proposed system is operationally feasible as it is aligned with organizational needs and supports improved operational performance.

Technical Feasibility

The technical feasibility of the TriOPS system is supported by a robust and professional technology stack specifically selected to handle the high volume operational data of Tagum Trio Lumber Corporation. The system will be developed as a modern web-based application, utilizing React Native and Tailwind CSS for the frontend to ensure a responsive and intuitive user experience across the company’s existing desktop and laptop hardware. To power the system's logic and server-side operations, Laravel (a PHP Framework) will be employed as the backend, providing a secure and efficient bridge between the user interface and the data layer. At the core of the technical infrastructure is a centralized MySQL database, which will serve as a secure and scalable repository for all mission critical information, including employee records, real-time production logs, and QR code scan data for product tracking and department transfers. This professional-grade database architecture is essential for the Workforce Analytics Platform providing the accurate integrated datasets required to generate the real-time dashboards and reports that will replace current manual paper-based processes. Given the team's technical expertise in these specific languages and the availability of the required hardware and networking tools, the development and deployment of the TriOPS system are highly feasible.

Table 2 Business Existing Technology

Technology	Particulars and Specification	Quantity
Software	Web Browser(Google Chrome/Microsoft Edge)	**8**
	Microsoft Excel	**1**
	Biometrics System Software	**1**
	Operating System(Windows)	**8**
Hardware	Printer	**2**
	Biometrics Device	**3**
	Desktop Computers	**8**
Network	Internet Connection	**1**
	Local Network	**1**

Stakeholder Analysis

Stakeholder analysis aims to evaluate and understand stake-holders from the perspective of an organization, or to deter-mine their relevance to a project or policy. In carrying out the analysis, questions are asked about the position, interest, influ-ence, interrelations, networks and other characteristics of stakeholders, with reference to their past, present positions and future potential [25] This study determines each stakeholder's level of participation, significance, and urgency to support informed decision-making and overall project success.

Table 3 Stakeholder Analysis

Stakeholder Name	**Contact Person** *Phone, Email, Website, Address*	**Impact** *How much does the project impact them? (Low, Medium, High)*	**Influence** *How much influence do they have over the project? (Low, Medium, High)*	**What is important to the stakeholder?**	**How could the stakeholder contribute to the project?**	**How could the stakeholder block the project?**	**Strategy for engaging the stakeholder**
Project Adviser	Stacey Nicole Marie Monta	High	High	Successful project completion and high academic quality standards.	Guides, validates outputs, and ensures quality standards	Withholding approval or rejecting outputs may significantly delay the project timeline	Schedule regular consultations and submit outputs promptly for review
Project Manager	Lance Benjamin C. Castillo	High	High	Delivering the project within the defined scope, timeline, and quality expectations	Plans, coordinates, and ensures timely delivery	Ineffective planning, poor coordination, or missed deadlines may compromise overall project delivery	Hold regular meetings and provide consistent progress updates
Project Developer	John Kier B. Turco	High	High	Developing a fully functional, efficient, and technically sound system	Designs, builds, and deploys the system based on defined	Submission of incomplete or defective system components may obstruct	Maintain clear communication and monitor development progress regularly

					project requirements	development progress	
Project Analyst	Ralph Zyron A. Namuag	High	Medium	Establishing accurate system requirements and a well-structured design	Identifies and documents system requirements to ensure the design meets project goals	Inaccurate or insufficient requirements gathering may lead to flawed system design and implementation	Collaborate closely during requirements gathering and design validation
General Manager	Jang Kuanrong	Medium	Medium	Obtaining reliable reports that support informed and strategic decision-making	Approves major project decisions and ensures alignment with organizational objectives	Withdrawal of support or redirection of organizational priorities may halt project advancement	Present updates formally and seek approval on major decisions
HR Personnel	Charlie Dela Cruz	Medium	Medium	Achieving streamlined HR operations through accurate and efficient data management	Provides relevant HR data and validates the accuracy of HR-related system features	Provision of inaccurate or incomplete HR data may impede the development of HR-related system features	Coordinate regularly to collect HR data and validate system features
Production Head	Rhejane Austral	High	Medium	Ensuring timely, accurate, and well-organized production data management	Supplies production data requirements and confirms the system meets operational needs	Failure to supply accurate production data or validate outputs may disrupt system development	Meet regularly to gather data needs and confirm system outputs
Production	Mr. Yen	Medium	High	Maintaining smooth	Shares operational	Limited availability for	Involve in system testing

Supervisor				operational workflow supported by a dependable system	knowledge and actively participates in system testing	testing or insufficient operational input may slow system refinement	and gather operational feedback consistently
Employees	Employees of Tagum Trio Lumber Corporation	High	High	Accessing an intuitive system that simplifies tasks and accelerates request processing	Engages in user testing and provides practical feedback to improve system usability	Reluctance to participate in testing or provide meaningful feedback may affect system usability and quality	Conduct testing sessions and encourage feedback on system usability
Accounting and Finance	Jamie Lafuente	Medium	Medium	Ensuring accurate financial records, reliable data integrity, and efficient processing of financial transactions within the system	Supplies financial data requirements and verifies the accuracy of financial system modules	Incomplete financial data or failure to validate modules may delay the completion of financial system components	Coordinate closely to gather financial data and validate modules

e. Project Charter

Every project begins with a solid plan, and a project charter is the document that makes that plan official. It captures the essential details of a project including its purpose, objectives, scope, the people involved, the resources available, the milestones to be achieved, the risks to be managed, and the approvals required before development proceeds. Table 5 presents the project charter developed by the proponents for TriOPS, which acted as the team's primary guide in ensuring that the system was built with clear direction, proper coordination, and a strong commitment to addressing the production monitoring, salary management, and workforce analytics needs of Tagum Trio Lumber Corporation.

TriOPS: Web-Based QR Production Process with Salary Management System and Workforce Analytics in Tagum Trio Lumber Corporation	
Project Description	The TriOPS: Web-Based QR Production Process with Salary Management System and Workforce Analytics for Tagum Trio Lumber Corporation is a web-based system designed to streamline the organization's production monitoring, salary management, and workforce analytics operations.
Project Objectives	The project aims to develop and implement TriOPS — a web-based QR Production Process with Salary Management System and Workforce Analytics that will centralize workforce information, automate core production processes, enable QR code-based production output scanning, support accurate piece rate-based salary monitoring, and assist management in making data-driven workforce decisions. Specifically, the system will provide modules for QR code generation and scanning, employee records management, employee reassignment management, production output monitoring, piece rate salary monitoring, deduction processing, workforce analytics, decision support dashboards, and report generation for Tagum Trio Lumber Corporation.
Project Considered Successfully When:	Project Considered Successfully When: The project is considered successful when TriOPS is fully developed, tested, and deployed for Tagum Trio Lumber Corporation; the major system modules are functional and aligned with the approved requirements; employee records, reassignment histories, and production output data can be managed through the system; QR code scanning accurately captures and records each employee's production output per piece; piece rate-based salary can be monitored and viewed by management at any time; workforce analytics, decision support dashboards, and reports can assist management in making informed, data-driven decisions on production performance and workforce operations; users including production supervisors, HR staff, and management are trained to operate the system; and the project is completed within the approved scope, schedule, and available resources.

Project Participants	**Project Adviser**: Ms. Stacey Nicole Marie Monta **Project Manager:** Mr. Lance Benjamin C. Castillo **Project Analyst:** Mr. Ralph Zyron A. Namuag **Project Developer**: Mr. John Kler B. Turco **Domain Expert / HR Representative**:Ms. Rhejane Austral **Client / Organization:** Tagum Trio Lumber Corporation **End Users:**General Manager , Accounting and Finance Personnel,Production Supervisor, Production In-Charge ,Production Leadman , Production Employees
Available Resources	The project will use available organizational resources such as desktop computers, laptops, printer, internet connection, local network, and QR code scanners for production output recording. The system will be developed using React.js, Tailwind, Python, FastAPI, Visual Studio Code, Postgre, web browsers, and other development tools. The project will also use approved documentation, existing HR forms, production output records, employee records, piece rate computation references, reassignment logs, process information, and stakeholder feedback as references for system development. The estimated overall project budget is ₱578,593.81, including manpower, audit, and technology costs.
Milestone	**Develop an Overall Model:**February 5, 2026- March 14, 2026 **Build a Feature List:**March 16, 2026-April 26,2026 **Plan by Feature:**April 9,2026-June 25,2026 **Design Core Features:**July 1, 2026-July 27, 2026 **Build by Feature:**July 29,2026-April 1,2027
Potential Risk	Possible risks include incomplete or inaccurate production output data from QR code scanning, resistance to system adoption by production workers and supervisors, delays in requirements validation, system bugs or technical errors, unstable internet or local network connection, incorrect employee reassignment encoding, data privacy and security issues, and possible changes in piece rate computation rules or organizational requirements during development. These risks will be addressed through user training, system testing, data validation, backup procedures, role-based access control, regular consultation with management and production supervisors, and continuous feedback during development
Approval	**Project Adviser:** Ms. Stacey Nicole Marie Monta Signature: ___________ Date: ___________ **Project Manager:** Mr. Lance Benjamin C. Castillo

	Signature: ___________ Date: ___________ **Domain Expert / HR Representative:** _____________________________ Signature: ___________ Date: ___________ **Client Representative / Management:** Ms. Rhejane Austral Signature: ___________ Date: ___________

f. Project Team Roles and Responsibilities

This section describes the roles and responsibilities of each project team member involved in the development of the TriOPS: Web-Based QR Production Process with Salary Management System and Workforce Analytics. It also includes the roles of end-users and stakeholders, who play an important role in the planning, evaluation, validation, and successful implementation of the system.

Table 4 Project Team Roles and Responsibilities

Role	Responsibilities
Project Manager	Responsible for planning, scheduling, and monitoring the overall progress of the project. Ensures that the project is completed within the required scope, time, and quality standards while facilitating communication among team members and stakeholders.
Project Adviser	Provides academic and technical guidance throughout the project development. Reviews and validates project outputs and recommends improvements for the system and documentation quality.
System Analyst	Responsible for gathering and analyzing system requirements. Studies existing processes and converts them into system specifications, models, and design documents.
System Developer	Responsible for developing the system, including coding, database design, integration, and testing. Ensures that the system is functional, reliable, and secure.
Production Personnel	Provides system requirements based on actual production processes and operations. Participates in testing and validation to ensure that the system properly supports production tracking, salary management, and workforce. analytics workflows.

g. Work Breakdown Structure

a. Work Breakdown Structure

The Work Breakdown Structure (WBS) for TriOPS: Web-Based QR Production Process with Salary Management System and Workforce Analytics for Tagum Trio Lumber Corporation provides a comprehensive division of the project into five major phases, sub-phases, and specific activities. Adopting the Feature-Driven Development (FDD) methodology, the structure outlines the development process starting from Developing an Overall Model, Building a Feature List, Planning by Feature, Designing by Feature, up to Building by Feature—which covers system construction, testing, deployment, and maintenance. This systematic framework ensures that all project components are properly organized, executed, and monitored from the initial planning stage through to the successful implementation and long-term operation of the system.

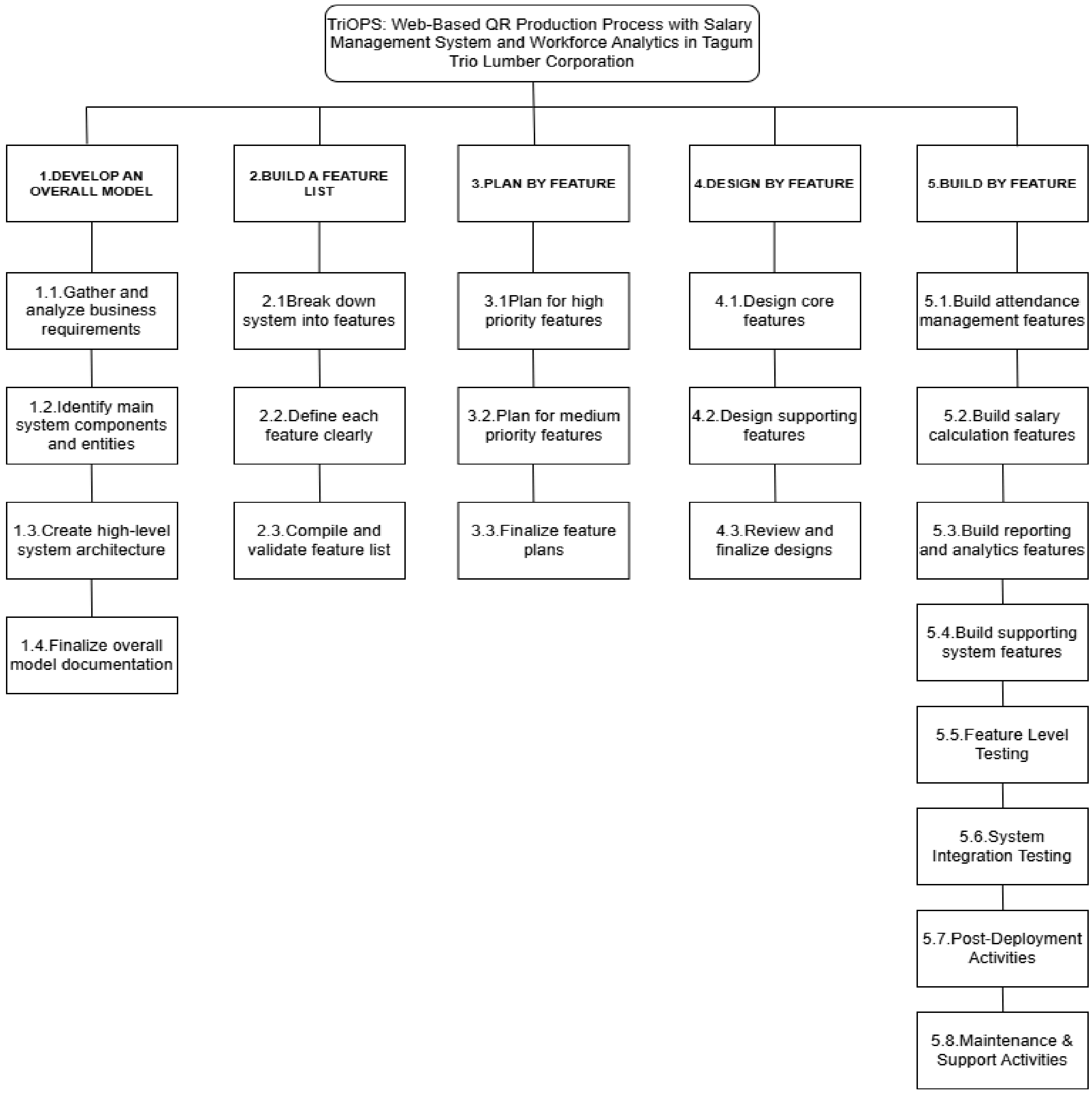


Figure 12 Work Breakdown Structure

Each work package defined in the Work Breakdown Structure (WBS) is further elaborated in the WBS Dictionary. This document provides clear descriptions of every activity, facilitating proper resource allocation, efficient task management, and accurate progress monitoring. It ensures that all outputs are completed in strict accordance with the project requirements of TriOPS: Web-Based QR Production Process with Salary Management System and Workforce Analytics for Tagum Trio Lumber Corporation.

WBS Dictionary

Table 6 presents the Work Breakdown Structure (WBS) Dictionary of TriOPS: Web-Based QR Production Process with Salary Management System and Workforce Analytics for Tagum Trio Lumber Corporation. The WBS Dictionary provides a detailed description of each work package, including its task identification number, task description, estimated man-days, start date, end date, and persons involved. It serves as a planning and monitoring guide for the proponents to ensure that all activities are properly scheduled, assigned, and completed according to the project requirements. The schedule follows the Modified Feature Driven Development (FDD) Model with Continuous Feedback Loop across all phases of development. Although the work packages are arranged according to the FDD structure — from Develop an Overall Model through Build by Feature — selected activities overlap to reflect the iterative and feature-driven nature of the development process. This allows feature design, development, testing, refinement, deployment preparation, and user validation to proceed in a coordinated manner without following a strictly linear sequence. The revised schedule also excludes Saturday and Sunday as scheduled start and end dates, ensuring that the timeline is realistic, manageable, and feasible for completion within the approved project duration, covering a total of 127.5 mandays from February 2026 to April 2027.

The following are the meanings of the abbreviated task owner:

PM – Project Manager

SDEV – System Developer

SA – System Analyst

Table 7. Work Breakdown Structure (WBS) Dictionary

Task ID	Task Description	Mandays 127.5	Start Date	End Date	Persons Involved
1	**DEVELOP AN OVERALL MODEL**	**18.5**	1-Feb-26	14-Mar-26	
1.1	**Define project scope and objectives**	**8**	5-Feb-26	13-Feb-26	**PM/SA**
1.1.1	Identify business goals for HR system	4	5-Feb-26	8-Feb-26	PM/SA
1.1.2	Determine system purpose and expected benefits	4	10-Feb-26	13-Feb-26	PM/SA
1.2	**Gather and analyze business requirements**	**8**	**18-Feb-26**	**27-Feb-26**	**SA**
1.2.1	Conduct initial interviews with stakeholders	3	18-Feb-26	20-Feb-26	SA
1.2.2	Review existing processes and documents	2	23-Feb-26	24-Feb-26	SA
1.2.3	Document user needs and functional requirements	3	25-Feb-26	27-Feb-26	SA
1.3	**Identify main system components and entities**	**2.5**	**4-Mar-26**	**10-Mar-26**	**SA**
1.3.1	Define core data entities: employee, attendance, salary, reports	0.5	4-Mar-26	4-Mar-26	SA
1.3.2	Establish relationships between entities	2	9-Mar-26	10-Mar-26	SA
1.4	**Create high-level system architecture**	**2**	**11-Mar-26**	**12-Mar-26**	**SA/SDEV**
1.4.1	Design system structure and modules	2	11-Mar-26	12-Mar-26	SA/SDEV

1.5	**Finalize overall model documentation**	**2**	**13-Mar-26**	**14-Mar-26**	**PM/SA**
1.5.1	Review and approve overall project model	2	13-Mar-26	14-Mar-26	PROJECT TEAM
2	**BUILD A FEATURE LIST**	**13.5**	**16-Mar-26**	**8-Apr-26**	**SA/SDEV**
2.1	**Break down system into features**	**8**	**16-Mar-26**	**25-Mar-26**	**SA/SDEV**
2.1.1	Identify attendance management features	3	16-Mar-26	18-Feb-26	SA/SDEV
2.1.2	Identify salary calculation features	2	19-Mar-26	20-Mar-26	SA/SDEV
2.1.3	Identify reporting and analytics features	3	23-Mar-26	25-Mar-26	SA/SDEV
2.2	**Define each feature clearly**	**5.5**	**26-Mar-26**	**3-Apr-26**	**SA**
2.2.1	Describe purpose and functionality of each feature	0.5	26-Mar-26	26-Mar-26	SA
2.2.2	Specify inputs, outputs and rules for each feature	2	30-Mar-26	31-Mar-26	SA
2.2.3	Classify features by priority and importance	3	1-Apr-26	3-Apr-26	SA/PM
2.3	**Compile and validate feature list**	**3**	**6-Apr-26**	**8-Apr-26**	**SA/SDEV**
2.3.1	Organize features in logical order	3	6-Apr-26	8-Apr-26	SA
3	**PLAN BY FEATURE**	**23.5**	9-Apr-26	26-Jun-26	**SA/SDEV**
3.1	**Plan for high priority features**	**11**	9-Apr-26	12-May-26	**SA**

3.1.1	Define scope and boundaries for each feature	2	9-Apr-26	10-Apr-26	DEV/PM
3.1.2	Estimate effort and time required	0.5	13-Apr-26	13-Apr-26	DEV
3.1.3	Identify required resources and skills	3	15-Apr-26	17-Apr-26	DEV/PM
3.1.4	Define dependencies between features	2	20-Apr-26	21-Apr-26	DEV/SA
3.1.5	Assign tasks to team members	3	5-May-26	7-May-26	PM
3.1.6	Create schedule and milestones	0.5	12-May-26	12-May-26	PM
3.2	**Plan for medium priority features**	**12.5**	**1-Jun-26**	**22-Jun-26**	
3.2.1	Outline implementation approach	3	1-Jun-26	3-Jun-26	DEV
3.2.2	Determine technical requirements	3	4-Jun-26	6-Jun-26	DEV/SA
3.2.3	Estimate costs and risks	3	10-Jun-26	12-Jun-26	PM/SA
3.2.4	Set delivery timelines	3	15-Jun-26	17-Jun-26	PM
3.2.5	Align with overall project plan	0.5	22-Jun-26	22-Jun-26	PM/SA
3.3	**Finalize feature plans**	**2**	25-Jun-26	25-Jun-26	**PM/SA**
3.3.1	Review all feature plans	2	25-Jun-26	26-Jun-26	PROJECT TEAM
4	**DESIGN BY FEATURE**	**19**	1-Jul-26	28-Jul-26	SA/SDEV

4.1	**Design core features**	**12**	1-Jul-26	16-Jul-26	SA/SDEV
4.1.1	Design user interface for attendance recording	3	1-Jul-26	3-Jul-26	SDEV
4.1.2	Design logic for attendance calculation	3	6-Jul-26	8-Jul-26	SDEV/SA
4.1.3	Design salary structure and rules	2	9-Jul-26	10-Jul-26	SDEV/HR
4.1.4	Design calculation formulas and conditions	2	13-Jul-26	14-Jul-26	SDEV/SA
4.1.5	Design basic report templates	2	15-Jul-26	16-Jul-26	SDEV/SA
4.2	**Design supporting features**	**5**	17-Jul-26	24-Jul-26	**SDEV**
4.2.1	Design data storage and database structure	2	17-Jul-26	18-Jul-26	SDEV/SA
4.2.2	Design data entry and validation rules	0.5	20-Jul-26	20-Jul-26	SDEV
4.2.3	Design access control and user roles	2	21-Jul-26	22-Jul-26	SA
4.2.4	Design system navigation and flow	0.5	24-Jul-26	24-Jul-26	SDEV
4.3	**Review and finalize designs**	**2**	27-Jul-26	28-Jul-26	**SDEV**
4.4.1	Check design against requirements and standards	2	27-Jul-26	28-Jul-26	PROJECT TEAM
5	**BUILD BY FEATURE**	**53.5**	29-Jul-26	2-April-27	**SDEV**
5.1	**Build attendance management features**	**6**	29-Jul-26	5-Aug-26	**SDEV**

5.1.1	Develop QR code generation and scanning function	2	29-Jul-26	30-Jul-26	SDEV
5.1.2	Develop time-in and time-out recording logic	1	31-Jul-26	31-Jul-26	SDEV
5.1.3	Develop attendance record storage and retrieval	1	3-Aug-26	3-Aug-26	SDEV
5.1.4	Develop attendance summary view	2	4-Aug-26	5-Aug-26	SDEV
5.2	**Build salary calculation features**	**7**	10-Aug-26	28-Aug-26	**SDEV**
5.2.1	Develop salary component setup interface	2	10-Aug-26	11-Aug-26	SDEV
5.2.2	Develop basic pay and deduction calculations	2	12-Aug-26	13-Aug-26	SDEV
5.2.3	Develop overtime and allowance calculations	1	19-Aug-26	19-Aug-26	SDEV
5.2.4	Develop final pay computation function	2	27-Aug-26	28-Aug-26	SDEV
5.3	**Build reporting and analytics features**	**4.5**	31-Aug-26	4-Sep-26	**SDEV**
5.3.1	Develop standard report generation	2	31-Aug-26	1-Sep-26	SDEV
5.3.2	Develop data visualization and charts	2	2-Sep-26	3-Sep-26	SDEV
5.3.3	Develop custom report creation and export	0.5	4-Sep-26	4-Sep-26	SDEV
5.4	**Build supporting system features**	**6.5**	4-Sep-26	12-Sep-26	**SDEV**

5.4.1	Develop user management and access control	0.5	4-Sep-26	4-Sep-26	SDEV
5.4.2	Develop data backup and maintenance tools	3	7-Sep-26	9-Sep-26	SDEV
5.4.3	Develop system settings and configuration	3	10-Sep-26	12-Sep-26	SDEV
5.5	**Feature Level Testing**	**6**	18-Sep-26	6-Oct-26	**SDEV**
5.5.1	Test individual features for functionality	2	18-Sep-26	19-Sep-26	SA/SDEV
5.5.2	Validate features against requirements	2	1-Oct-26	2-Oct-26	SDEV/SA
5.5.3	Document defects and issues	2	5-Oct-26	6-Oct-26	SA
5.6	**System Integration Testing**	**6**	7-Oct-26	6-Nov-26	**SDEV**
5.6.1	Test interaction between all modules	2	7-Oct-26	8-Oct-26	SDEV
5.6.2	Verify data flow and consistency	2	2-Nov-26	3-Nov-26	SDEV
5.6.3	Check system compatibility and performance	2	5-Nov-26	6-Nov-26	SDEV
5.7	**Post-Deployment Activities**	**6.5**	22-February -27	5-March-27	**PROJECT TEAM**
5.7.1	Conduct user training sessions	3	22-February -27	24-February -27	SDEV
5.7.2	Prepare user manuals and documentation	0.5	2-March-27	2-March-27	SA
5.7.3	Provide initial technical support	3	3-March-27	5-March-27	SDEV

5.8	**Maintenance & Support Activities**	**11**	8-March-27	2-April-27	**SDEV**
5.8.1	Monitor system performance and usage	2	8-March-27	9-March-27	SDEV
5.8.2	Perform regular data backups	2	15-March-27	16-March-27	SA
5.8.3	Apply updates, patches and security fixes	2	17-March-27	18-March-27	SDEV
5.8.4	Address user issues and requests	3	23-March-27	25-March-27	SDEV
5.9.5	Document maintenance activities and progress	2	1-April-27	2-April-27	SA

f. Gantt Chart

The Gantt chart presented in Figure 13 illustrates the project schedule for TriOPS: Web-Based QR Production Process with Salary Management System and Workforce Analytics for Tagum Trio Lumber Corporation. It outlines the timeline of activities following the five phases of the Feature-Driven Development (FDD) methodology, as defined in the Work Breakdown Structure (WBS). The project commences with the Develop an Overall Model phase from February to March 2026, where system requirements and architecture are established. This is followed by the Build a Feature List and Plan by Feature phases, scheduled from March to April 2026, focusing on defining, organizing, and prioritizing system functionalities. The Design by Feature phase runs from April to July 2026, covering the detailed design of core and supporting system features. The longest phase, Build by Feature, extends from July 2026 through March 2027; this stage includes system development, feature-level and integration testing, post-deployment implementation, and maintenance activities such as bug fixes, performance monitoring, and continuous improvements based on user feedback. The Gantt chart ensures that all project activities are systematically scheduled, properly monitored, and aligned with the defined WBS, budget, and overall project objectives.

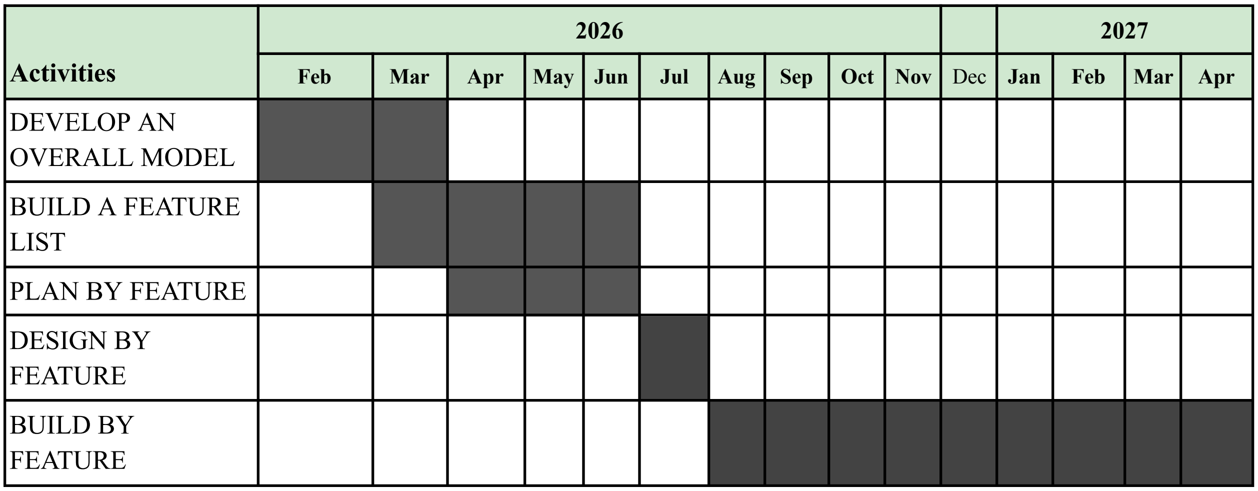


Figure 13. Gantt Chart

B. System Requirements and Analysis

a. Budget Summary

b. Budget Allocation and Schedule

The Budget Allocation and Schedule outlines the systematic distribution of project costs from February 2026 to March 2027, aligned with the implementation timeline. Audit Cost remains consistent at ₱1,000.00 per month throughout the entire duration of the project. Effort Cost, which covers manpower expenses based on the Work Breakdown Structure (WBS) activities, gradually increases from ₱5,000.00 in the initial months to ₱9,000.00 from July 2026 onward, before tapering down as the project reaches completion. Technology Cost is kept minimal, primarily applied during key phases—set at ₱1,200.00 from March 2027 onwards to support deployment and maintenance needs. The Total Cost represents the combined monthly expenses for technology, audit, and effort, while the Cumulative Cost tracks the progressive sum of all expenditures, reaching a final overall project cost of ₱115,500.00 upon completion.

Table 7. Budget Allocation and Schedule

YEAR	MONTH	Technology Cost	Audit Cost	Effort Cost	Total Cost	Cumulative Cost
2026	February		₱1,000.00	₱5,000.00	₱6,000.00	₱6,000.00
	March		₱1,000.00	₱5,000.00	₱6,000.00	₱12,000.00
	April		₱1,000.00	₱6,000.00	₱7,000.00	₱19,000.00
	May		₱1,000.00	₱6,000.00	₱7,000.00	₱26,000.00
	June		₱1,000.00	₱6,000.00	₱7,000.00	₱33,000.00
	July		₱1,000.00	₱8,000.00	₱9,000.00	₱42,000.00
	August		₱1,000.00	₱8,000.00	₱9,000.00	₱51,000.00
	September		₱1,000.00	₱8,000.00	₱9,000.00	₱60,000.00
	October		₱1,000.00	₱8,000.00	₱9,000.00	₱69,000.00
	November		₱1,000.00	₱8,000.00	₱9,000.00	₱78,000.00
	December		₱1,000.00	₱8,000.00	₱9,000.00	₱87,000.00

2027	**January**		₱1,000.00	₱6,000.00	₱7,000.00	₱94,000.00
	February	₱2,100.00	₱1,000.00	₱6,000.00	₱9,100.00	₱103,100.00
	March	₱1,200.00	₱1,000.00	₱5,000.00	₱7,200.00	₱110,300.00
	April	₱1,200.00	₱1,000.00	₱3,000.00	₱5,200.00	₱115,500.00
Total		**₱4,500.00**	**₱15,000.00**	**₱96,000.00**	**₱115,500.00**	**₱115,500.00**

Effort Cost [Ralph Task]

Hardware Cost [Ralph Task]

Hardware	**Quantity**	**Unit Price**	**Total Cost**
Desktop Computer	**4**	₱ 18,500/each	Available
Cellphone	**11**	₱ 5,000/each	Available
Printer	**1**	₱ 8,000	Available

Cash Outlay [Ralph Task]

Software Cost [Ralph Task]

Software	Quantity	Unit Price	Total Cost
VS Code			
Postgre			
Fast API			
React.js			

c. Financial Viability[Ralph Task]

d. Cost and Benefit Description

Process	Current Process (Minutes)	Proposed System (Minutes)	Time Saved (Minutes)
Production Recording	35	15	20
Salary Computation	60	20	40
Report Generation	30	15	15
Payroll Processing	40	25	15
TOTAL	**165**	**75**	**90**

Table ___ presents the comparison between the current manual processes and the proposed TriOPS system in terms of processing time. Based on the analysis of the existing and proposed workflows, TriOPS significantly reduces the time required to complete the major production and salary-related operations of Tagum Trio Lumber Corporation.

Production recording, which currently takes 35 minutes through manual logging and paper-based tracking of lumber output, is reduced to only 15 minutes through the QR code-based production scanning feature of TriOPS — resulting in a time savings of 20 minutes. Salary computation, which is currently performed manually by computing each employee's earnings based on the piece, thickness, and size of finished lumber products, is reduced from 60 minutes to 20 minutes through the system's automated piece rate salary computation feature, saving 40 minutes of processing time. Report generation, which currently requires manual collation and preparation of production and salary data, is reduced from 30 minutes to 15 minutes through the system's automated report generation module, saving 15 minutes. Payroll processing, which currently takes 40 minutes due to manual review and preparation of salary records, is reduced to 25 minutes through the system's salary monitoring and summary generation features, resulting in a time savings of 15 minutes.

Overall, the total processing time is reduced from 165 minutes to 75 minutes, resulting in a total time savings of 90 minutes. This demonstrates that TriOPS significantly improves operational efficiency, reduces the manual workload of HR personnel and production supervisors, and supports faster and more accurate salary computation and production monitoring for Tagum Trio Lumber Corporation.

Table 15. Expenses: Current Process

Table 16. Expenses: Proposed Process

Table 17. Cost Comparison: Current vs Proposed Process

Table 18. Estimated Cost Savings

Table 19. Cumulative Cost vs. Cumulative Savings

e. RACI Chart

Table ___ presents the RACI Chart of TriOPS: Web-Based QR Production Process with Salary Management System and Workforce Analytics for Tagum Trio Lumber Corporation. The RACI Chart defines the roles and responsibilities of each project team member across all tasks and activities within the five phases of the

Modified Feature Driven Development (FDD) model.

In this project, the Project Manager is generally accountable for ensuring that all project activities are completed according to the approved scope, schedule, and requirements of TriOPS. The System Developer is responsible for all technical activities including system design, database development, feature development, unit testing, system integration testing, deployment, and system maintenance of TriOPS. The System Analyst is responsible for requirements analysis, process documentation, system modeling, feature identification and prioritization, testing preparation, and validation support throughout the five phases of the Modified FDD Model. The End Users composed of the HR Personnel/Admin, Production Supervisors, and General Manager of Tagum Trio Lumber Corporation provide input regarding existing production processes and salary computation practices, validate system requirements, participate in User Acceptance Testing (UAT), and support the implementation and deployment of TriOPS. They are also informed and actively involved in selected activities, particularly during feature validation, user testing, training, and feedback gathering to ensure that the system remains aligned with the actual operational needs of Tagum Trio Lumber Corporation.

Table 20. RACI Chart

LEGEND	
R – Responsible	The person(s) who does the work to complete the task.
A – Accountable	The person who is ultimately answerable for the correct and thorough completion of the task.
C – Consulted	The people who provide input or advice related to the task. These are usually subject matter experts.
I – Informed	The people who are kept up to date on progress or decisions but are not directly involved in doing the work.

TriOPS: Web-Based QR Production Process with Salary Management System and Workforce Analytics in Tagum Trio Lumber Corporation				
Main Activity / Task	**Project Manager**	**System Analyst**	**System Developer**	**EndUsers**

1. DEVELOP OVERALL MODEL				
Define project scope, goals, and boundaries	A	R	C	C
Gather business context and domain knowledge	C	R	C	A
Create high-level system model and structure	A	R	C	I
Define architecture and technical framework	A	R	C	I
Approve the Overall Model	A	C	I	C
2. BUILD FEATURE LIST				
Identify all required functionalities/features	C	R	C	A
Describe features clearly (Action → Result → Object)	C	R	C	C
Group features by business area or module	A	R	C	I
Prioritize features based on value and risk	A	R	I	C
Finalize and sign-off the complete Feature List	A	C	I	R
3. PLAN BY FEATURE				
Estimate effort, complexity, and timeline	A	R	C	I
Identify dependencies and sequence of work	A	R	C	I
Assign tasks and resources to the team	A	C	R	I
Create development schedule and milestones	A	R	C	I
Confirm delivery dates with stakeholders	A	C	I	R
4. DESIGN BY FEATURE				
Detail functional requirements and specifications	C	R	C	C
Create workflow diagrams and process logic	C	R	A	I
Design database structure and system components	C	A	R	I
Prepare technical and user documentation drafts	I	R	A	I
Review and approve design package	A	R	C	C

5. BUILD BY FEATURE				
Write code and build components per design	C	C	R	I
Perform unit testing and code reviews	C	R	A	I
Integrate components and perform system tests	C	R	A	I
Fix defects and refine features	C	C	R	I
Conduct User Acceptance Testing (UAT) support	C	R	C	A
Final review and approval for deployment	A	C	R	I
Deploy and deliver the working feature	A	C	R	I

f. System Analysis

Before a system can be developed, it is essential to first establish a clear and comprehensive set of requirements that will serve as the foundation of the entire development process. System requirements refer to the specific functionalities, capabilities, and constraints that TriOPS must fulfill in order to properly address the operational needs of Tagum Trio Lumber Corporation. These requirements were gathered and derived through careful analysis of the corporation's existing business processes, day-to-day workflows, and the actual needs of its users — covering key areas such as QR code-based production output scanning, piece rate salary monitoring, employee records management, employee reassignment, and workforce analytics and reporting. To organize these requirements in a structured and meaningful way, they are classified into two categories: functional requirements, which specify the core features and capabilities that the system must perform, and non-functional requirements, which define the quality standards, performance expectations, security measures, and usability constraints that the system must consistently meet throughout its operation.

System Requirements

Functional Requirements

The functional requirements describe the specific features and operations that TriOPS must perform. The proposed Web-Based QR Production Process with Salary Management System and Workforce Analytics shall provide the following functions:

1. The system shall allow the HR/Admin to store, view, add, update, and manage employee profiles, supporting documents, and employment details within Tagum Trio Lumber Corporation.
2. The system shall allow the HR/Admin to reassign employees to different production areas or work assignments, and maintain a complete history of all employee reassignments for tracking and reference purposes.
3. The system shall allow Leadman production to scan QR codes to record and capture the quantity of production output completed by each worker per piece.
4. The system shall allow the HR Personnel/Admin/Production In-Charge and authorized users to view, monitor, and manage all recorded production output data of employees at any time through the system.
5. The system shall automatically compute each employee's salary based on the total recorded production output and the corresponding rate assigned according to the piece, thickness, and size of each lumber product completed by the employee.
6. The system shall allow authorized users to view and monitor each employee's computed piece rate salary, production performance, and earnings summary at any given time.
7. The system shall allow the Admin and Production In-Charge to set up, update, and manage piece rate values assigned to specific production tasks or product types within Tagum Trio Lumber Corporation.
8. The system shall maintain a complete and accurate record of each employee's production output history, piece rate earnings, and salary summaries over time for monitoring and reference purposes.
9. The system shall allow authorized users to generate, view, and export production output summaries and salary computation records for a specific employee, production area, or time period.
10. The system shall restrict access to functions, records, and data based on user roles such as HR Personnel/Admin, Production Supervisor, General Manager, and other authorized personnel of Tagum Trio Lumber Corporation.

11. The system shall maintain a system log that records all user activities, including logins, data entries, updates, deletions, and report generations, for accountability and audit purposes.
12. The system shall provide workforce analytics that display production performance trends, salary summaries, output comparisons per employee or production area, and other key metrics to assist management in making informed and data-driven workforce decisions.
13. The system shall send system-generated notifications or alerts to authorized users when significant changes are made to employee records, production output entries, piece rate values, or salary computations.
14. The system shall allow authorized users to back up and restore system data to prevent data loss and ensure the continuity of operations in Tagum Trio Lumber Corporation.

Non-Functional Requirements

The non-functional requirements define the performance, security, usability, reliability, and maintainability characteristics that TriOPS must consistently meet throughout its operation. The proposed system shall satisfy the following requirements:

1. The system shall provide a clean, intuitive, and user-friendly interface with straightforward navigation, clearly labeled controls, and organized layouts to ensure that HR Personnel/Admin, Production Supervisors, General Managers, and all other authorized users of Tagum Trio Lumber Corporation can operate the system with ease, even without extensive technical training.
2. The system shall ensure the accuracy, completeness, and consistency of all stored data, including employee profiles, production output records, piece rate configurations, salary computation results, employee reassignment histories, workforce analytics outputs, and system activity logs.
3. The system shall process all user interactions, including QR code scanning inputs, production output recording, piece rate salary computations, workforce analytics generation, decision support processing, and report creation, within a

reasonable response time that does not interrupt or delay the daily production operations of Tagum Trio Lumber Corporation.

4. The system shall enforce secure user authentication through unique login credentials and password protection, and shall implement role-based access control to ensure that each user can only access the functions and records permitted by their assigned role within the organization.
5. The system shall protect all sensitive data including employee personal information, production output records, piece rate values, salary summaries, and workforce analytics results from unauthorized access, modification, or disclosure through appropriate data protection and encryption measures.
6. The system shall follow a modular and feature-driven design structure consistent with the Modified FDD Model used in the development of TriOPS, allowing individual modules to be updated, debugged, enhanced, or replaced without disrupting the functionality of the entire system.
7. The system shall be scalable and capable of accommodating a growing volume of employee records, production output entries, piece rate configurations, salary computation histories, reassignment logs, and analytics data as the workforce and production operations of Tagum Trio Lumber Corporation expand over time.
8. The system shall be fully accessible through widely used web browsers such as Google Chrome and Microsoft Edge on desktop computers and laptops, ensuring compatibility with the existing hardware infrastructure of Tagum Trio Lumber Corporation without requiring additional software installation.
9. The system shall support scheduled and on-demand data backup procedures to safeguard all critical records, including employee information, production output data, piece rate values, salary summaries, reassignment histories, and system logs, and shall allow full data recovery in the event of accidental deletion, system failure, or data corruption.
10. The system shall guarantee that all piece rate salary computations, production output tallies, workforce analytics calculations are precise, consistent, and entirely based on the verified production output records, configured piece rate

values, and analytics parameters defined by authorized users — eliminating the risk of manual computation errors.

11. The system shall be deployable and accessible through a local area network (LAN) or internet connection depending on the network infrastructure and deployment preferences of Tagum Trio Lumber Corporation, ensuring flexibility in how the system is hosted and accessed across different work areas.
12. The system shall maintain a detailed and tamper-evident system activity log that automatically records all user actions, including logins, logouts, data entries, modifications, deletions, report generations, and configuration changes, to support accountability, transparency, and audit compliance within the organization.
13. The system shall provide a well-structured, documented, and maintainable codebase and database architecture that enables the development team and future system administrators to efficiently perform troubleshooting, apply updates, and implement system enhancements in response to the evolving operational needs of Tagum Trio Lumber Corporation.
14. The system shall display clear and informative feedback messages to users when actions are successfully completed, when errors are encountered, or when invalid inputs are detected, ensuring that users are always informed of the system's current status and can take appropriate corrective action without confusion.
15. The system shall be designed to minimize downtime and ensure continuous availability during regular working hours of Tagum Trio Lumber Corporation, so that General Manager production supervisors, HR personnel,Accounting Staff and management can access and use the system whenever needed without interruption
16. The system shall enforce data integrity constraints at both the application and database levels to prevent duplicate records, incomplete entries, and inconsistent data from being saved — particularly for critical records such as employee profiles, QR code assignments, production output entries, and piece

rate configurations.

Process Model: Use Case Diagram and Description

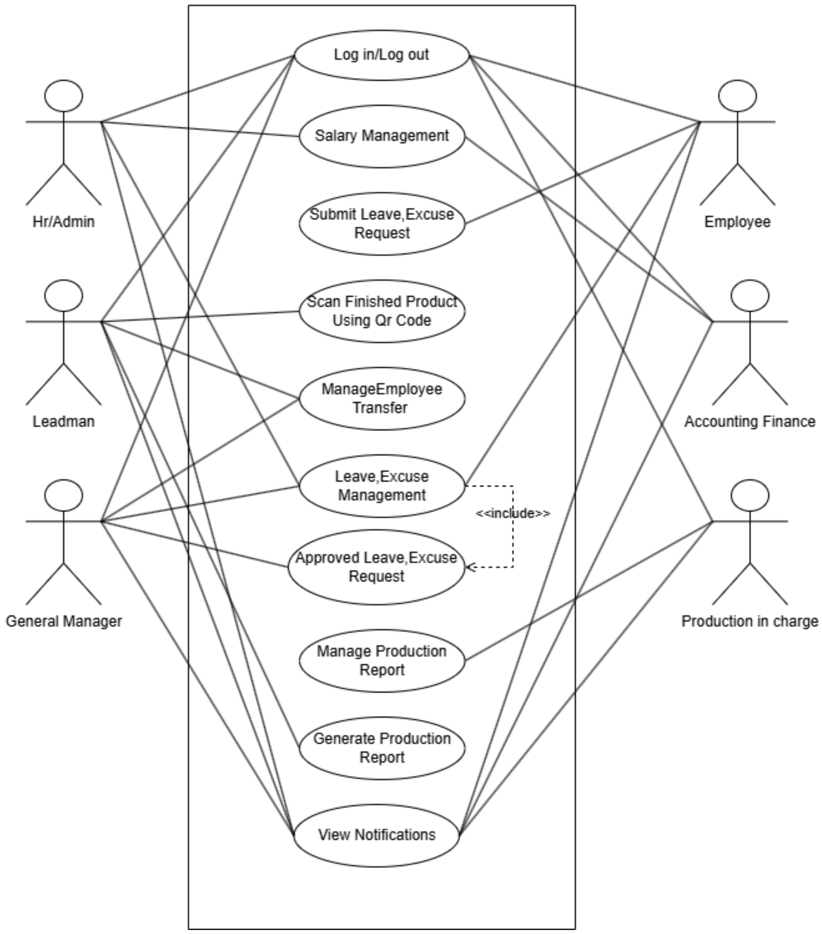


Figure Use Case Diagram and Description

To fully understand how TriOPS functions from the perspective of its users, use case descriptions are provided to give a comprehensive and detailed walkthrough of each major system function presented in the Use Case Diagram. Every use case outlines the specific actors involved, the conditions that must be met before the process begins, the step-by-step flow of events during system interaction, the alternative or exception scenarios that may occur, the conditions that signal the end of the process, the possible risks associated with each function, and the quality standards that the system must meet. These descriptions serve as a clear and structured reference for understanding exactly how the HR Personnel/Admin, Production Supervisors, General Manager, Leadman,Production Supervisor, Production In-Charge, and other authorized personnel of Tagum Trio Lumber Corporation interact with TriOPS on a day-to-day basis and how the system is designed to handle both expected operations and unexpected situations with accuracy and reliability.

The following table presents the complete and detailed use case descriptions of TriOPS: Web-Based QR Production Process with Salary Management System and Workforce Analytics for Tagum Trio Lumber Corporation.

Table 00 Use Case Description for Log in / Log Out

Use Case Name	Log in / Log Out
Participating Actors	• HR/Admin • General Manager • Leadman • Accounting Finance • Production Supervisor/ Production In-ChargeProduction In-Charge
Entry Conditions	The user has an existing account and needs to access the system.
Flow of Events	1. The user accesses the login page. 2. The user enters their username/email and password. 3. The system validates the entered credentials. 4. If valid, the system grants access based on the user's role. 5. The user performs the necessary system tasks. 6. The user clicks the log out button after completing the task. 7. The system ends the user session and redirects the user to the login page.

Alternative / Exception Scenarios	If the username/email or password is incorrect, the system displays an error message and denies access. If the account is inactive or unauthorized, the system prevents login and prompts the user to contact the HR Manager/Admin.
Exit Conditions	The user successfully logs in or logs out of the system.
Quality Requirements	The system must ensure secure authentication, role based access control, session protection, and proper logout functionality.

Table 00 Use Case Description for Salary Management

Use Case Name	Salary Management
Participating Actors	• HR/Admin • General Manager • Accounting Finance
Entry Conditions	The user must be logged in with an authorized role. Production output records must already be recorded in the system.
Flow of Events	1. The HR/Admin,Accounting Finance or authorized user navigates to the Salary Management module. 2. The system displays the list of employees with their corresponding recorded production output. 3. The system automatically computes each employee's salary based on the total recorded production output and

	the corresponding rate assigned according to the piece, thickness, and size of each lumber product completed. 4. The authorized user reviews the computed salary details of each employee. 5. The General Manager reviews and validates the salary summaries. 6. The system saves and updates the finalized salary records.
Alternative / Exception Scenarios	If no production output has been recorded for an employee, the system displays zero earnings for that employee. If a piece rate value has not been configured for a specific product, the system prompts the authorized user to set up the rate before the salary computation proceeds. If the system is unavailable or encounters an error during salary computation, the authorized user may manually calculate the salary based on the daily production report and encode the results into the system once it becomes available.
Exit Conditions	The salary computations are successfully calculated, reviewed, and saved in the system for all employees with recorded production output.
Quality Requirements	The system must ensure that all salary computations are accurate, consistent, and strictly based on verified production output records and configured piece rate values per piece, thickness, and size of lumber products.

Table 00 Use Case Description for Submit Leave, Excuse Request

Use Case Name	Submit Leave, Excuse Request
Participating Actors	• Employee
Entry Conditions	The employee must be logged in to their account in the TriOPS system.
Flow of Events	1. The employee navigates to the Leave / Excuse Request section of their dashboard. 2. The employee fills out the request form, specifying the type of request, date, and reason. 3. The employee submits the request through the system. 4. The system records the submission and sends a notification to the authorized approver. 5. The employee receives a confirmation that the request has been successfully submitted and is pending approval
Alternative / Exception Scenarios	If the employee submits an incomplete form, the system displays a validation error and prompts the employee to fill in all required fields before resubmitting. If the requested date conflicts with an existing approved request, the system notifies the employee of the conflict.
Exit Conditions	The leave or excuse request is successfully submitted and recorded in the system, and the approver has been notified.
Quality Requirements	The system must ensure that all submitted requests are accurately recorded, properly notified to approvers, and accessible for review at any time.

Table 00 Use Case Description for Scan Finished Product Qr Code

Use Case Name	Scan Finished Product using Qr Code
Participating Actors	• Leadman
Entry Conditions	The Leadman must be logged in to their account in the TriOPS system. Finished lumber products must be ready for recording.
Flow of Events	1. The Leadman navigates to the QR scanning function within the TriOPS system. 2. The Leadman scans the QR code of the finished lumber product using the available scanning device. 3. The system identifies the product associated with the scanned QR code, including its piece, thickness, and sizes. 4. The system automatically links the scanned finished product to the employee who produced it. 5. The system records and saves the production output under the identified employee's profile. 6. The system automatically calculates the salary of every employee under the Leadman based on their total recorded production output and the corresponding rate assigned according to the piece, thickness, and size of each finished lumber product. 7. The system updates the production output summary and salary computation of all affected employees and reflects the changes on the dashboard.
Alternative / Exception Scenarios	If the QR code of the finished product is unrecognized or invalid, the system displays an error message and prompts the Leadman to rescan or verify the product's QR code. If the QR code scanner is unavailable or not functioning, the

	Leadman may manually type and record the finished product details including the piece, thickness, and size directly into the system. If the input quantity or product details are incomplete, the system prevents the entry from being saved until all required fields are properly filled. If no piece rate value has been configured for the recorded product type, the system notifies the Leadman and prompts the authorized user to set up the rate before the salary computation proceeds.
Exit Conditions	The finished product output is successfully recorded under the correct employee's profile and the salary of all employees under the Leadman has been automatically computed and updated in the system.
Quality Requirements	The system must accurately capture and record all production output data per scan, ensure that each entry is correctly linked to the right employee, and guarantee that all salary computations are precise, consistent, and automatically updated based on verified production output records and configured piece rate values.

Table 00 Use Case Description for Manage Employee Transfer

Use Case Name	Manage Employee Transfer
Participating Actors	• The HR/Admin • General Manager
Entry Conditions	The HR/Admin or General Manager must be logged in with authorized access. The employee to be transferred must have

	an existing and active profile in the system.
Flow of Events	1. The General Manager opens the Production Page or scouts the available production departments to assess workforce distribution. 2. The General Manager identifies the department with the least number of employees and decides to transfer an employee to balance workload. 3. The employee is notified of the transfer through the system. 4. The employee accesses their account and submits a formal request for approval to transfer to the identified production department. 5. The Leadman of the receiving department receives a notification and accepts the reassigned employee. 6. The system records the transfer details, updates the employee's department assignment, and logs the reassignment history. 7. The Production In-Charge monitors the reassigned employee's production output in the new department. 8. Once the monitoring period is complete, a decision is made on whether to return the employee to their original department or keep them in the new one. 9. The system updates the final department assignment accordingly.
Alternative / Exception Scenarios	If the employee's transfer request is rejected, the system notifies the employee and retains their original department assignment. If no available slot exists in the target department, the system alerts the General Manager before proceeding.
Exit Conditions	The employee transfer is successfully recorded, the department assignment is updated in the system, and all

	involved parties have been notified of the final reassignment status.
Quality Requirements	The system must ensure that all transfer records are accurately documented, that reassignment histories are maintained and accessible, and that notifications are promptly delivered to all involved actors.

Table 00 Use Case Description for Leave, Excuse Management

Use Case Name	Leave, Excuse Management
Participating Actors	Employee, HR, General Manager
Entry Conditions	The General Manager must be logged in with authorized access. A leave or excuse request must have already been submitted by an employee.
Flow of Events	1. The HR navigates to the Leave / Excuse Management section of the dashboard and reviews all pending leave and excuse requests submitted by employees under their production area. 2. The HR evaluates the request based on the current production workload and endorses it to the General Manager for final approval. 3. The General Manager receives a notification regarding the endorsed request and navigates to the Leave / Excuse Management section. 4. The General Manager reviews the full details of the request including the type, date, and reason provided by the employee.

	5. The General Manager approves or rejects the request. 6. The system records the General Manager's decision, updates the status of the request, and notifies the employee of the outcome.
Alternative / Exception Scenarios	If a request contains incomplete or invalid information, the General Manager may reject it and indicate the reason for rejection. If the system detects a duplicate request from the same employee for the same date, it flags the conflict and notifies the Leadman for resolution before it is forwarded to the General Manager.
Exit Conditions	The leave or excuse request has been approved or rejected by the General Manager, and the employee has been notified of the final decision.
Quality Requirements	The system must ensure that all request statuses are updated accurately and promptly, that both the Leadman and General Manager can view and manage requests efficiently, and that all decisions made are properly recorded and traceable in the system.

Table 00 Use Case Description for Approved Leave, Excuse Management

Use Case Name	Approved Leave Excuse Management
Participating Actors	• General Manager
Entry Conditions	A leave or excuse request must have been reviewed and approved by the HR/Admin or General Manager through

	the Leave / Excuse Management function.
Flow of Events	1. The system processes the approved leave or excuse request. 2. The system automatically updates the employee's records to reflect the approved leave or excuse. 3. The employee receives a system notification confirming that their request has been approved. 4. The approved request is recorded in the system and becomes accessible for reference in reports and salary-related summaries.
Alternative / Exception Scenarios	If the notification fails to reach the employee due to a system error, the system logs the failed notification and retries delivery. If the approval was made in error, the HR/Admin may reverse the decision and update the record accordingly.
Exit Conditions	The approved leave or excuse request is fully recorded in the system and the employee has been successfully notified of the approval.
Quality Requirements	The system must ensure that all approved requests are accurately reflected in the employee's records, that notifications are reliably delivered, and that approved data is consistently available for reporting and salary computation references.

Table 00 Use Case Description for Manage Production Report

Use Case Name	Manage Production Report
Participating Actors	• Production In-Charge
Entry Conditions	The authorized user must be logged in. Production output data must already be recorded in the system
Flow of Events	1. The authorized user navigates to the Manage Production Report section of the dashboard. 2. The system displays all recorded production output data, organized by employee, production area, product type, and time period. 3. The authorized user reviews the production records and verifies the accuracy of the entries. 4. The authorized user may update, correct, or annotate production entries as necessary. 5. The system saves all changes and maintains an updated production record for reporting and salary computation purposes.
Alternative / Exception Scenarios	If a production entry is found to be incorrect or duplicated, the authorized user may flag or delete the entry and the system logs the action for audit purposes. If no production data is available for the selected period, the system displays an empty record and notifies the user accordingly.
Exit Conditions	All production records have been reviewed, verified, and updated in the system, and the data is ready for report generation and salary computation.
Quality Requirements	The system must ensure that all production records are complete, accurate, and properly organized to support reliable

	salary computation and workforce analytics for Tagum Trio Lumber Corporation.

Table 00 Use Case Description for Generate Production Report

Use Case Name	Generate Production Report
Participating Actors	• Leadman
Entry Conditions	The authorized user must be logged in. Production output records must be available and verified in the system.
Flow of Events	1. The authorized user navigates to the Generate Production Report section of the dashboard. 2. The user selects the desired report parameters such as employee, production area, product type, or time period. 3. The system processes the selected parameters and compiles the relevant production output data. 4. The system generates the production report and displays it on the screen. 5. The authorized user reviews the generated report. 6. The user may choose to print or export the report in the preferred format.
Alternative / Exception Scenarios	If no data matches the selected report parameters, the system notifies the user that no records are available for the specified criteria. If the report generation fails due to a system error, the system displays an error message and prompts the user to try again.
Exit Conditions	The production report is successfully generated, reviewed, and

	exported or printed by the authorized user.
Quality Requirements	The system must generate reports that are accurate, complete, and properly formatted, and must allow authorized users to export or print reports efficiently for management review and decision-making purposes.

Table 00 Use Case Description for View Notification

Use Case Name	View Notification
Participating Actors	● HR/Admin ● Leadman ● General Manager ● Employee ● Accounting Finance ● Production Supervisor / Production In-Charge
Entry Conditions	The user must be logged in to their account in the TriOPS system. At least one notification must have been generated by the system.
Flow of Events	1. The user logs in to their respective dashboard in TriOPS. 2. The system automatically displays the number of unread notifications on the dashboard. 3. The user clicks on the notification icon or section to view the list of notifications. 4. The system displays all notifications relevant to the user's role, such as transfer approvals, leave request updates, production output entries, salary computation results, and system alerts.

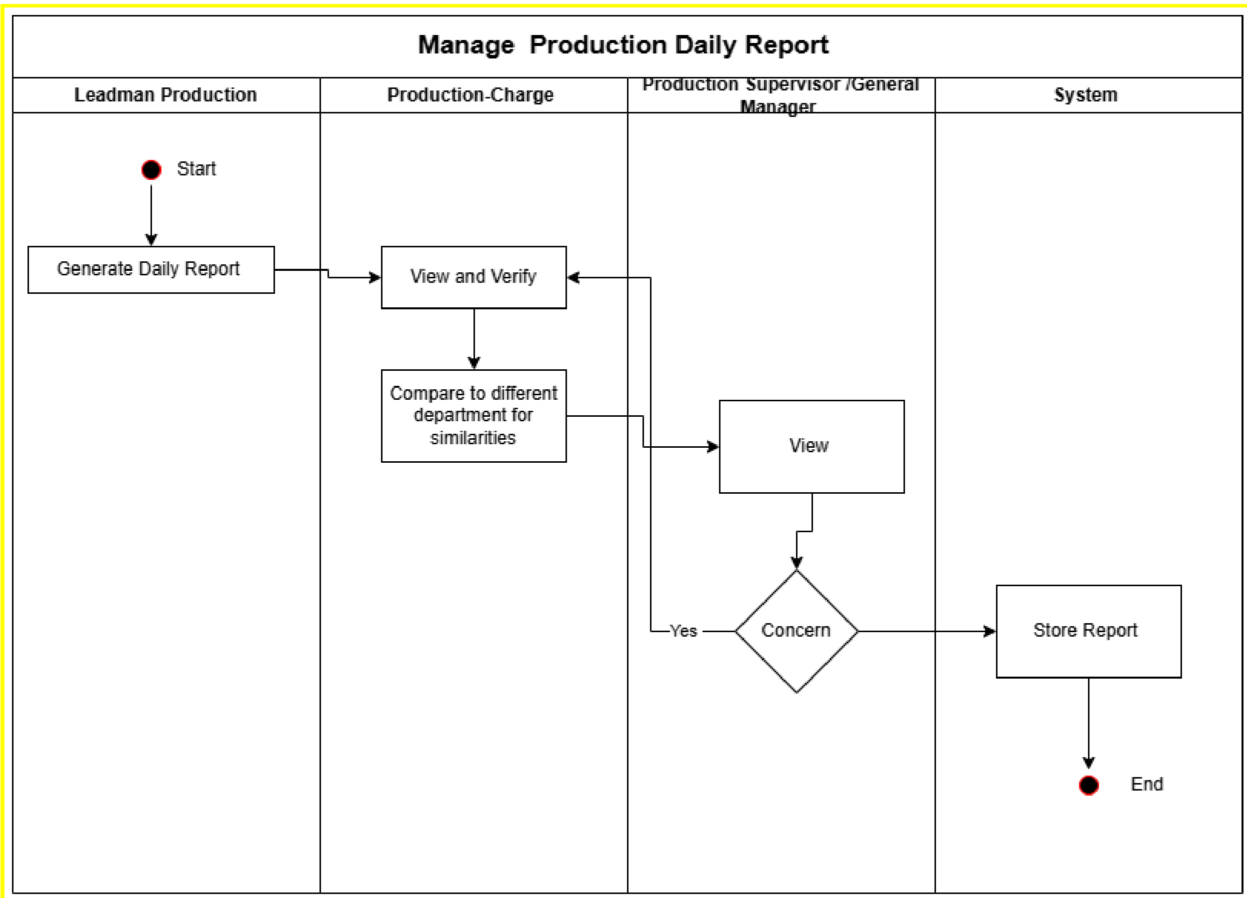
	5. The user reads the notification and takes the necessary action if required. 6. The system marks the notification as read and updates the notification count accordingly.
Alternative / Exception Scenarios	If there are no new notifications, the system displays a message indicating that there are no unread notifications at the moment. If a notification fails to load due to a system error, the system displays an error message and prompts the user to refresh the page.
Exit Conditions	The user has successfully viewed all relevant notifications and the system has updated the notification status accordingly.
Quality Requirements	The system must ensure that notifications are delivered promptly, displayed accurately based on the user's role, and properly marked as read once viewed to avoid confusion and ensure that no important updates are missed by any authorized user of Tagum Trio Lumber Corporation.

Process Model: Activity Diagram

The Process Model presents the step-by-step workflow of TriOPS using Activity Diagrams. These diagrams illustrate how each major process flows within the system — from the moment a user initiates an action to the point where the system completes its response. Each diagram captures the sequence of activities, key decision points, and the interactions among the HR Personnel/Admin, Leadman, General Manager, Employee, Accounting Finance, and Production In-Charge of Tagum Trio Lumber Corporation. Together, these diagrams provide a clear and structured visual representation of how tasks are carried out, how data moves through the system, and how TriOPS handles both normal operations and exceptional situations across its core functions — including QR-based production output recording, piece rate salary computation, employee transfer management, leave and excuse request processing, and workforce analytics reporting.

Activity Diagram for Scan Finished Product Qr Code

Activity Diagram for Production Report



Data Model: Level 1 Data Flow Diagram

C. System Design

a. Database Design: Entity-Relationship Diagram

b. Data Dictionary

c. System Architecture

D. System Development and Deployment

a. Test Cases, Alpha Testing, and UAT

Data Model: Context Data Flow Diagram

The Context Data Flow Diagram presents the overall flow of data between the TriOPS: Web-Based QR Production Process with Salary Management System and Workforce Analytics in Tagum Trio Lumber Corporation and its external entities. The system is represented as a single process labeled 0, indicating that it contains several internal functions such as QR production processing, salary management, workforce analytics, leave and attendance management, payroll computation, worker deployment, and report generation. The diagram shows the exchange of data among the Employee, Leadman Production, Production Supervisor, Production In-Charge, Accounting Head and Accounting Staff, Human Resource Personnel, General Manager, and the proposed system. The Employee receives leave and excuse status notifications, work schedule and section assignments, company announcements, and salary slips, while submitting excuse and absence notifications, reassigning requests, and leave applications to the system. The Leadman Production submits daily production log entries and QR production reports, while receiving QR scan error and validation alerts, department production target reports, worker output summaries, and reassigning confirmations. The Production Supervisor provides worker assignment per department and production concern and issue flags, and receives a production dashboard, QR scan summary reports, worker deployment reports, worker output rate comparisons, reassigned worker notifications, section production progress reports, and section task assignment logs. The Production In-Charge submits worker performance observations, department production data, employee record creation and updates, and policy and compliance updates to the system. The Accounting Head and Accounting Staff supply payroll summaries for final review, salary rate updates, and deduction and allowance entries, and receive deductions, payslip and salary computation reports, piece-rate computation breakdowns, and payroll audit trail logs in return. The Human Resource Personnel receives salary approval notifications, employee record and history reports, workforce analytics in HR view, and leave balance and attendance summaries from the system. The General Manager provides leave approval and denial decisions, department reassignment orders, salary approval confirmations, and company announcements, and receives workforce analytics reports, production performance summaries, pending request queues, and payroll summaries for final review.

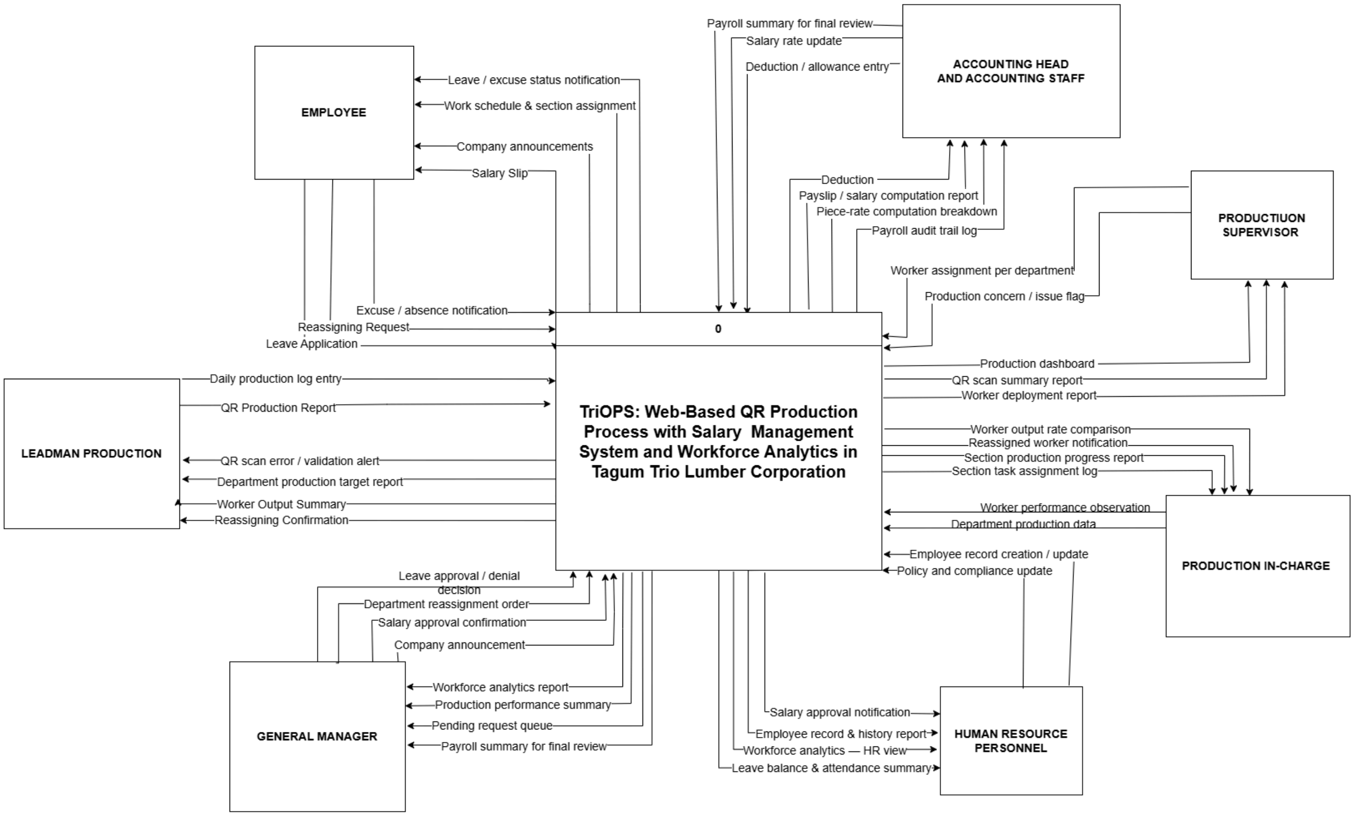


Figure 23. Context Data Flow Diagram

Reference

[5] A. . Fallis, “No Title No Title,” J. Chem. Inf. Model., vol. 53, no. 9, pp. 1689–1699, 2013.
[6] Işık, G., & Çifci, M. C. (2023). A Model Proposal for Scaling the Productivity Increase in Agile Project Management Methodology. International Journal of Pioneering Technology and Engineering, 2(02), 147-164.
[25]Lindenberg M, Crosby B. 1981. Managing development: the political dimension.

Hartford, CT: Kumarian Press.